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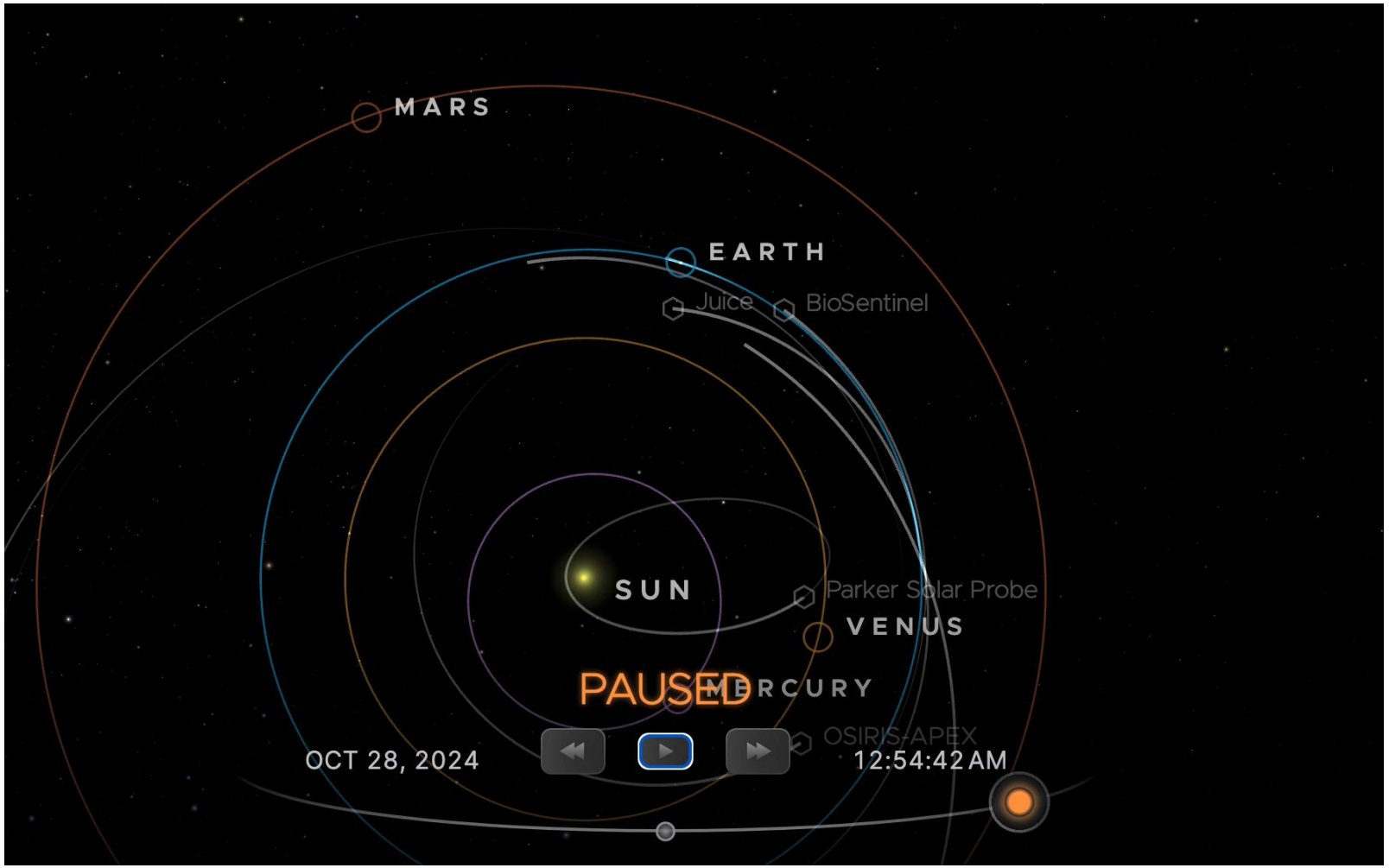
# Constraining Geophysical Properties of Europa's Ice Shell from the Depth of Eutectic Interfaces

**Defense for M.Sc. Planetary Science and Space Exploration**
**William Byrne**


**Supervisors**
Dr. Ana-Catalina Plesa
Prof. Dr. Frank Postberg

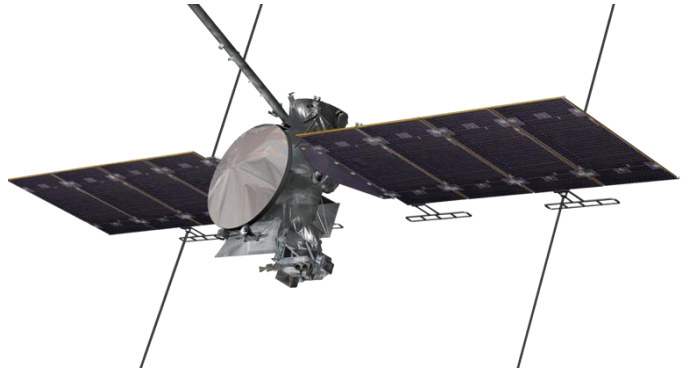

DLR Deutsches Zentrum für Luft- und Raumfahrt

# Relevant Missions
## Introduction



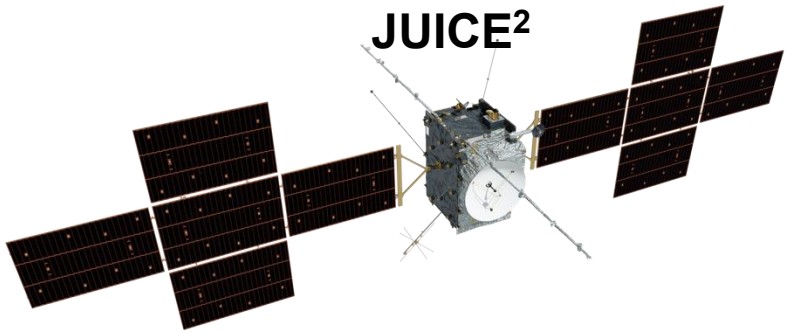

Source: https://eyes.nasa.gov/apps/solar-system

### Europa Clipper[1]



- Orbit Insertion: April 2030
- Orbital Focus: Europa

### JUICE[2]

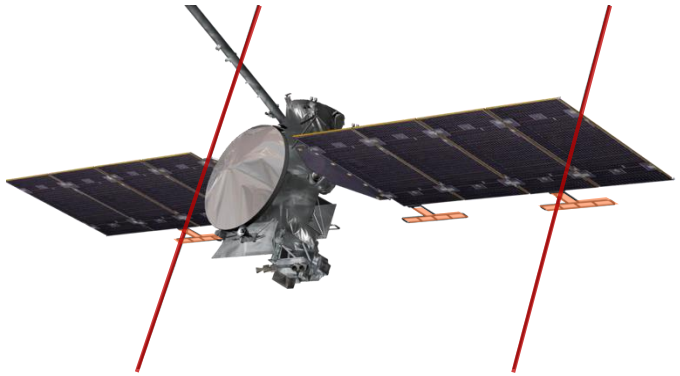


- Orbit Insertion: July 2031
- Orbital Focus: Ganymede

[1]jpl.nasa.gov, [2]sci.esa.int

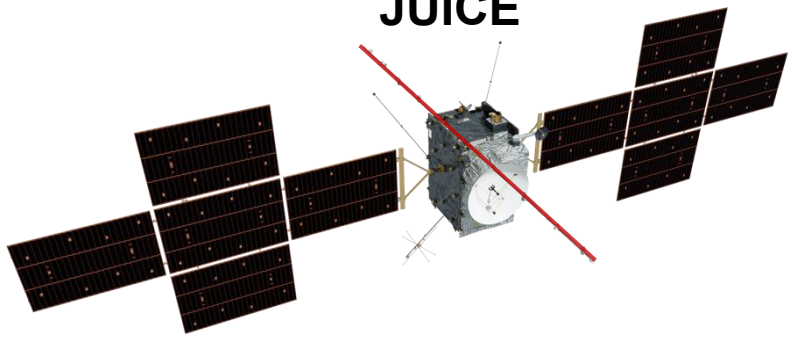
# Europa
## Introduction

**Europa Clipper**



- 49 planned flybys (most < 100 km)
- Radar Sounder: 9 MHz & 60 MHz

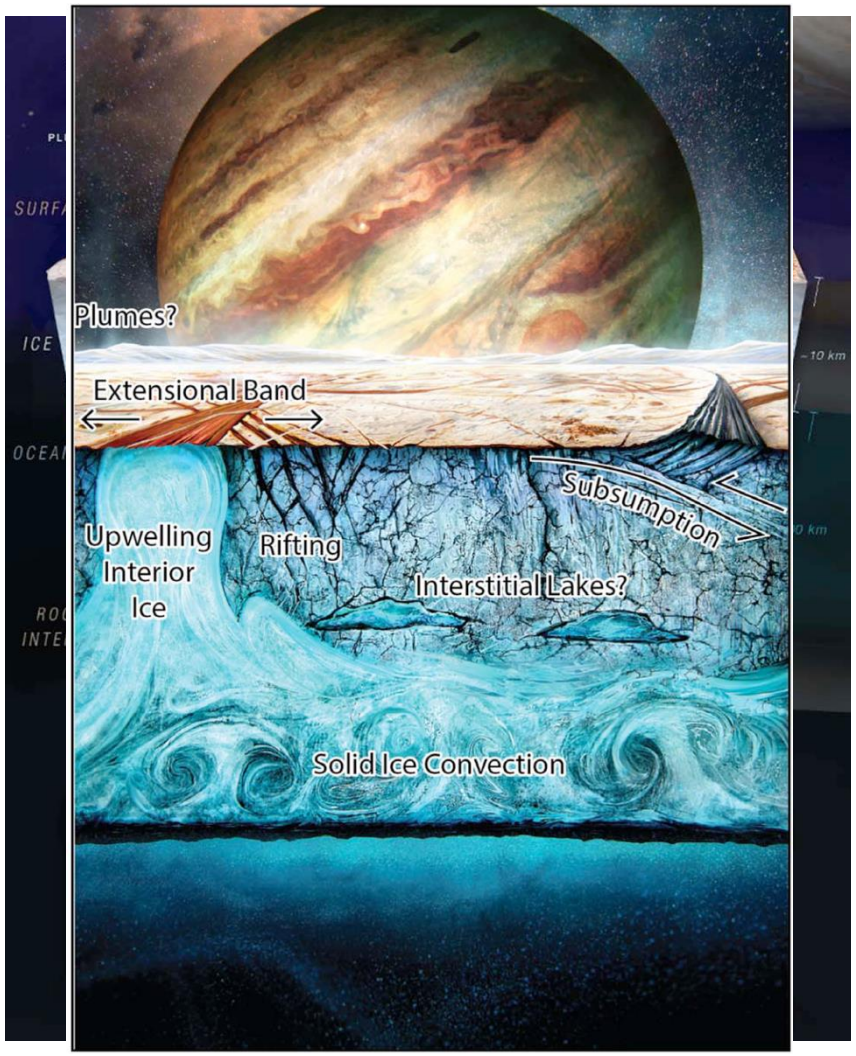
**JUICE**



- 2 planned flybys (~400 km)
- Radar Sounder: 9 MHz

Jupiter's Moon Europa

- Geologic features (lenticulae, chaos terrain)
- Surface composition (trailing/leading hemisphere)
- Formation theory (circumplanetary disk accretion)
- Internal structure modeling (ice shell dynamics)
- Induced magnetospheric modeling
- ....




Howell & Pappalardo, 2020

# Radar Sounding
## Introduction

### Radar Sounding

- Many radar signals (chirps) transmitted per second
- Return signals modified by dielectric gradients
- Used for decades in terrestrial glaciology

### Earth-based study

- Glacial depth (Ice-bed/ocean interface mapping)
- Fern/snow densification
- Sub-glacial water detection

Fig. 3. Airborne radar sounding system for ice sheets

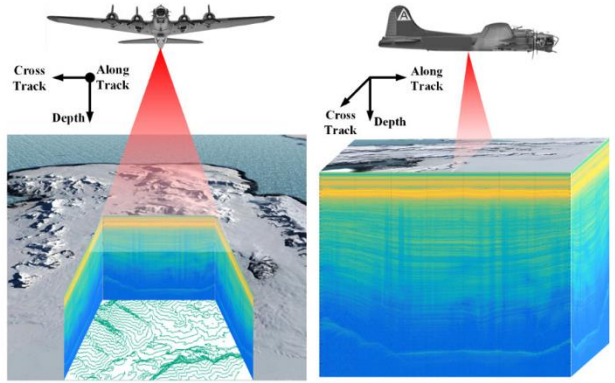
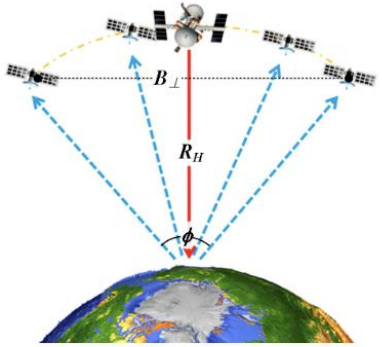
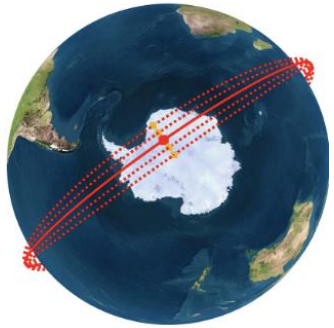



Fig. 4a. Cross-track multistatic radar.




(a)

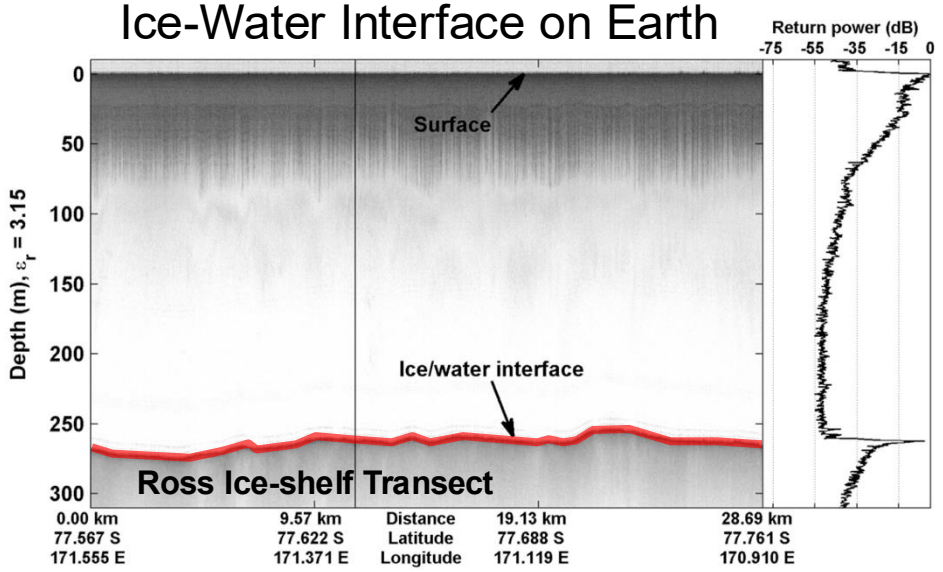
Fig. 4b. Collection of satellite orbits

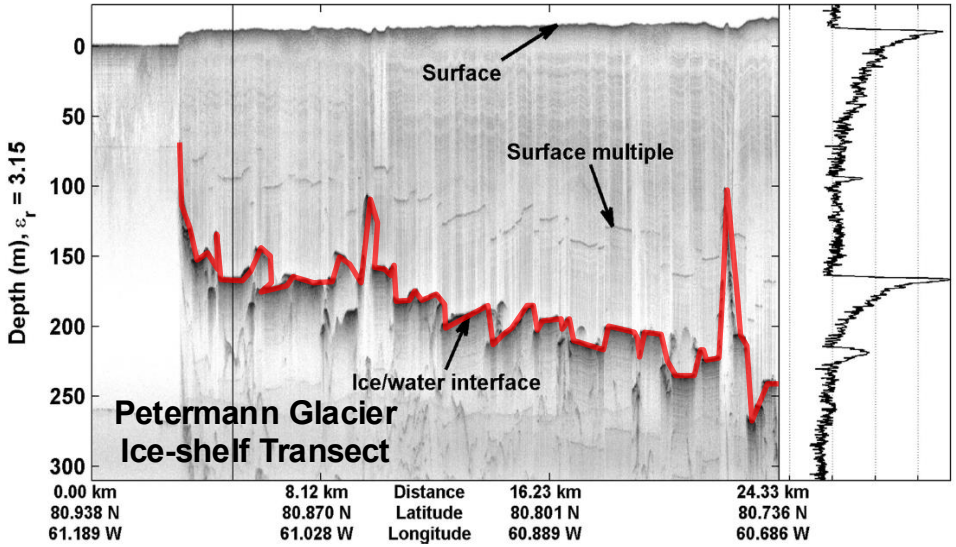


(b)

Xiao et al., 2022

Ice-Water Interface on Earth

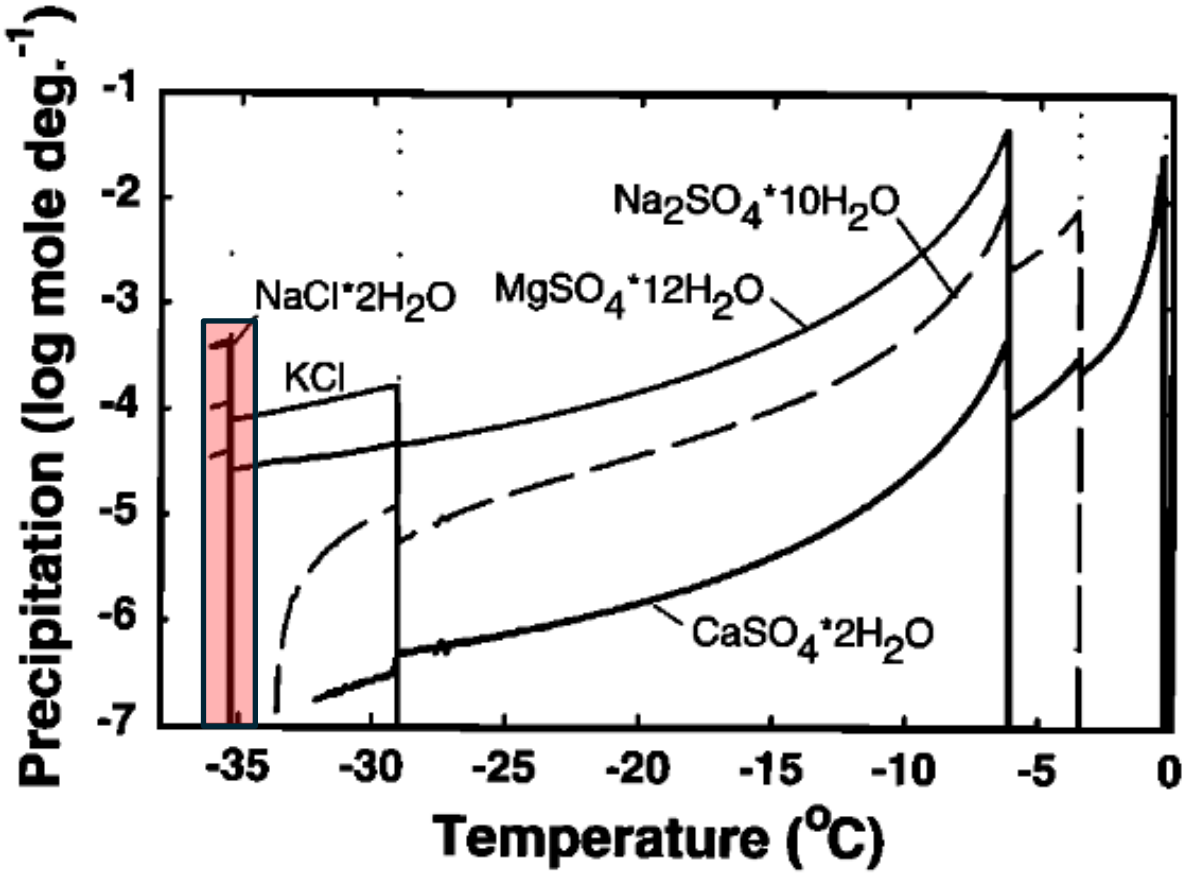



Lewis et al., 2015

# Eutectic Temperatures
## Introduction

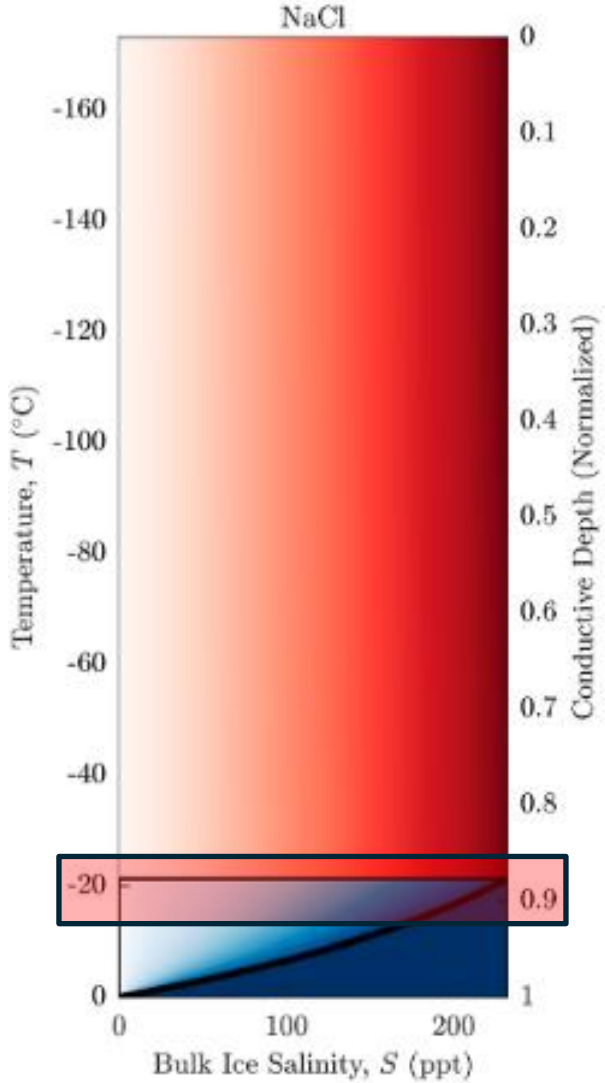
- "Compositionally-dependent temperature at which melt becomes thermodynamically stable" - **Souček et al. 2024**
- Context of Europa: The coldest temperature within the ice shell at which salty liquid water (brine) could form
- Shallower target for radar sounder than the ice-ocean interface

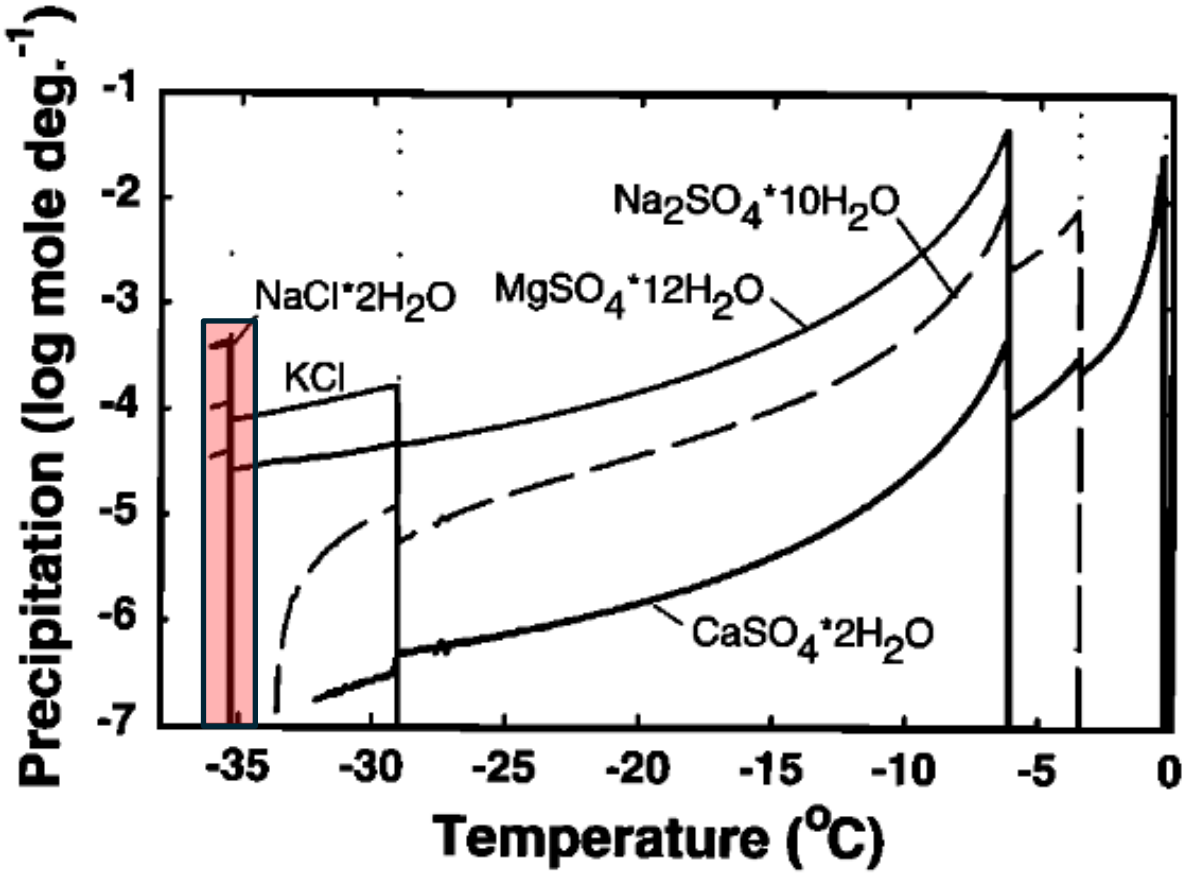



- **Zolotov & Shock, 2001**
- Mg–Ca–Na–K–Cl–$SO_4$–$H_2O$
- -36 °C, or **237.15 K**

# Eutectic Temperatures

## Introduction




- **Wolfenbarger et al., 2022**
- Binary end member: NaCl-$H_2O$
- -21.1°C, or **252.05 K**

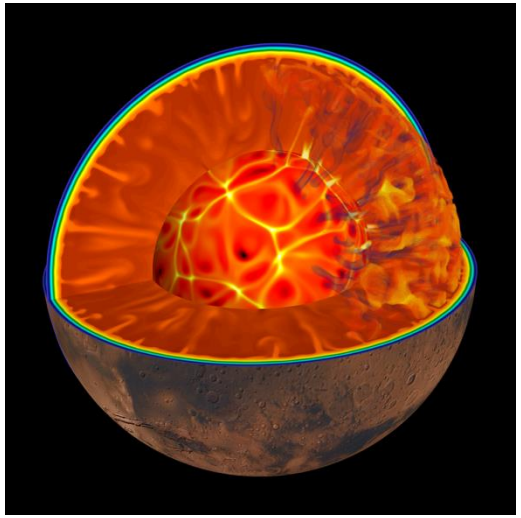
- **Zolotov & Shock, 2001**
- Mg–Ca–Na–K–Cl–$SO_4$–$H_2O$
- -36 °C, or **237.15 K**

# Geodynamics
## Introduction

**Stokes Flow**

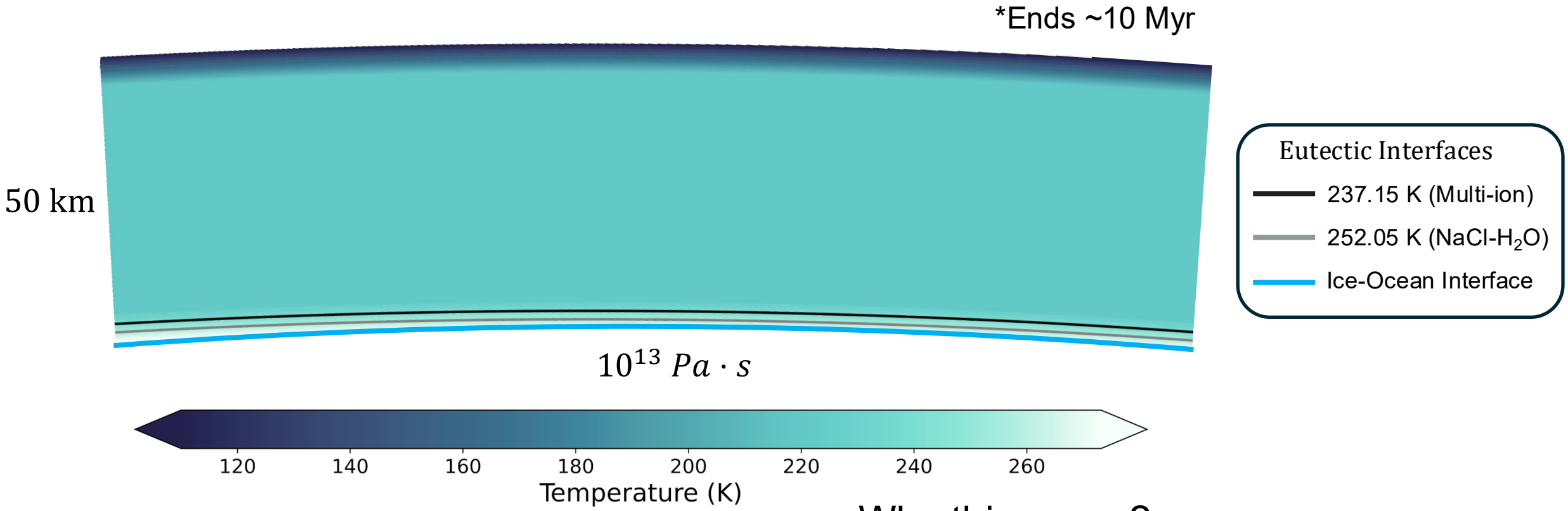
| | |
|---|---|
| **Conservation of Mass** | $\nabla \cdot \vec{u} = 0$ |
| **Conservation of Momentum** | $\nabla p - \nabla \cdot \left(\eta \left(\nabla \vec{u} + (\nabla \vec{u})^T\right)\right) = Ra\alpha T \vec{e}_r$ |
| **Conservation of Energy** | $\frac{\partial T}{\partial t} + \vec{u} \cdot \nabla T - Di\alpha(T + T_{surf})\vec{u_r} - \frac{Di}{Ra}\Phi = \nabla \cdot (k \nabla T)$ |

- Simulates fluid flow over geologic time scales
- Software: Generic Automation for planetary Interior Analysis, or GAIA **(Hüttig & Stemmer, 2008)**
- Track the depth to eutectic temperatures (**eutectic interface**)



**Christian Hüttig, DLR**

# Example Thermal Evolution
## Simulations




## Baseline Parametric Variation

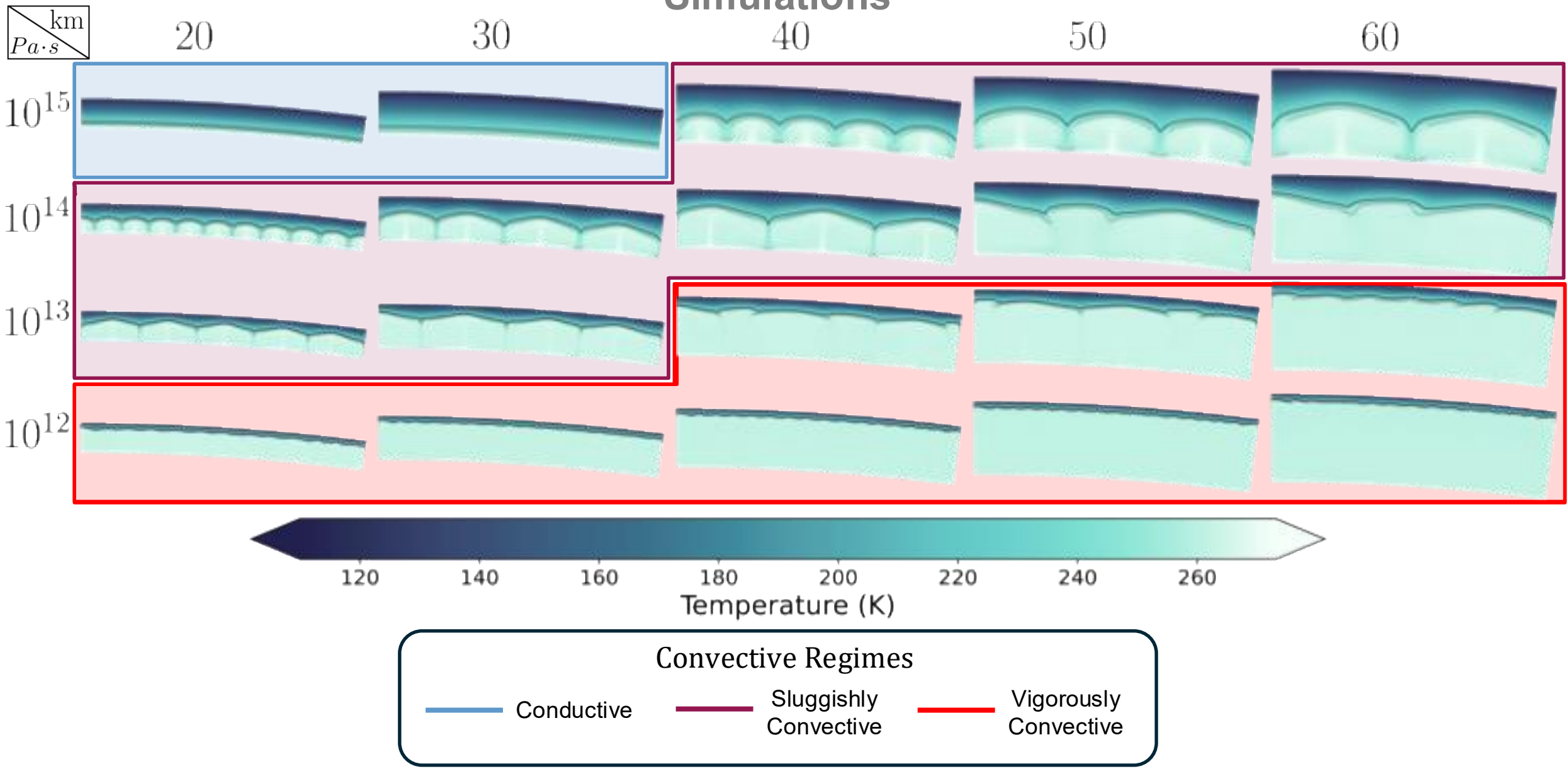
- Ice Shell Thickness (D): 20 - 60 km
- Reference Viscosity ($\eta_{ref}$): $10^{12}$–$10^{15}$ $Pa \cdot s$

## Why this range?

- Poorly constrained parameters
- Covers all possible convective regimes
- Affects isotherm depth & geometry

# Baseline Temperature Fields

## Simulations



# Scientific Questions

1. How detectable are these eutectic interfaces?

2. Can one use properties of eutectic interfaces to geophysically characterize Europa's ice shell?

# Radar Attenuation

## Lateral Detectability

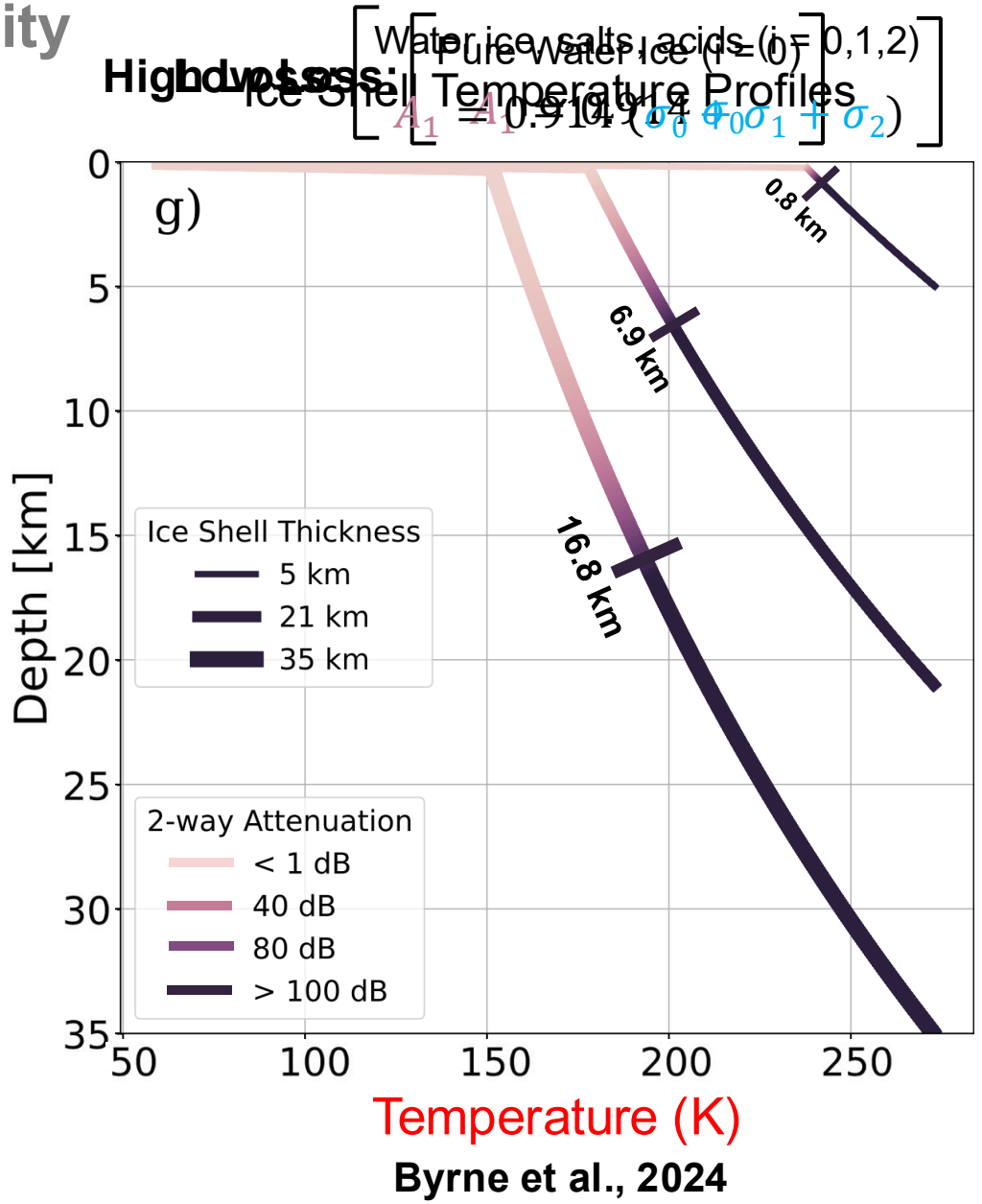
- Two-way Attenuation ($A_2$): Total Reduction in power (dB) as signal moves depth-wise (d) through material:

$$A_2(d) = \sum_{i=0}^{i_d} 2A_1(d_i)\Delta d$$

- Proportional to ice electrical conductivity ($\sigma$):

$$A_1 \approx 0.914\,\sigma \qquad \sigma = \sum_{i=0}^{2} \sigma_i C_i \exp\left[-\frac{E_i}{k_b}\left(\frac{1}{T} - \frac{1}{T_r}\right)\right]$$

- Estimated maximum attenuation limit for planetary radar sounders of **100 dB (Kalousová et al., 2017)**

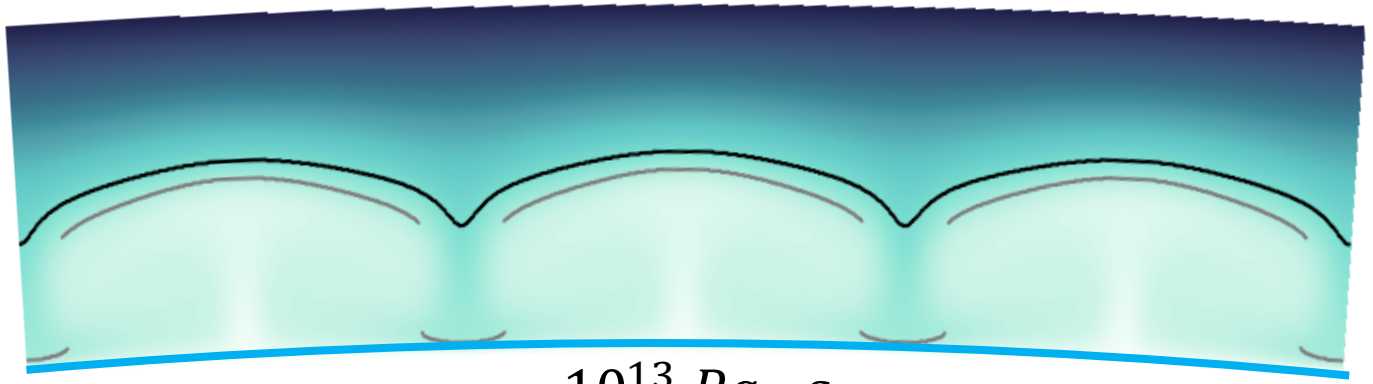


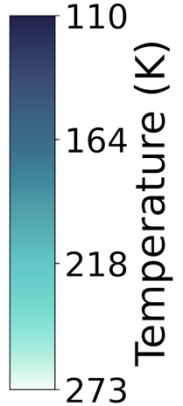

**Byrne et al., 2024**

means no more signal

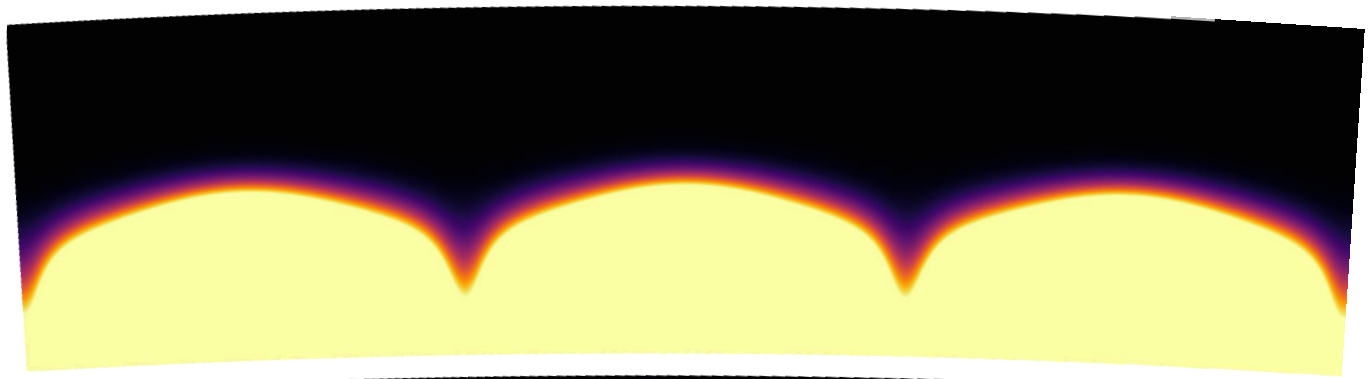
# 2D Radar Attenuation
## Lateral Detectability

50 km

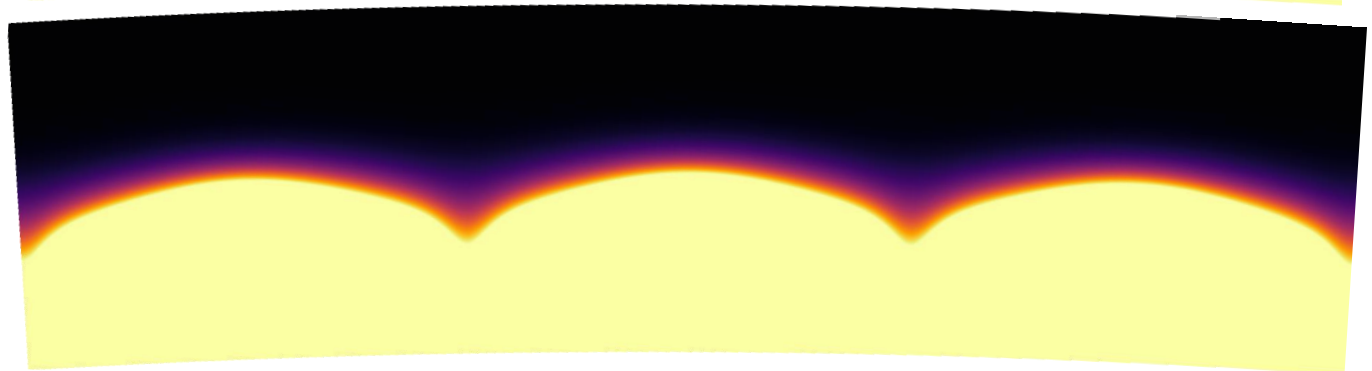


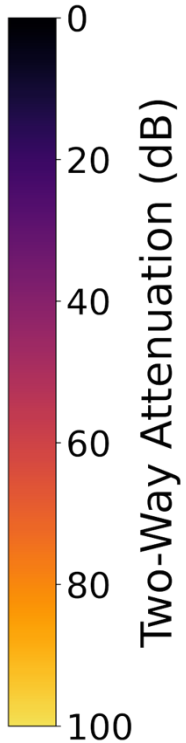




**Low Loss** Model
(Pure water Ice)



**High Loss** Model
(Ice + Salts + Acids)

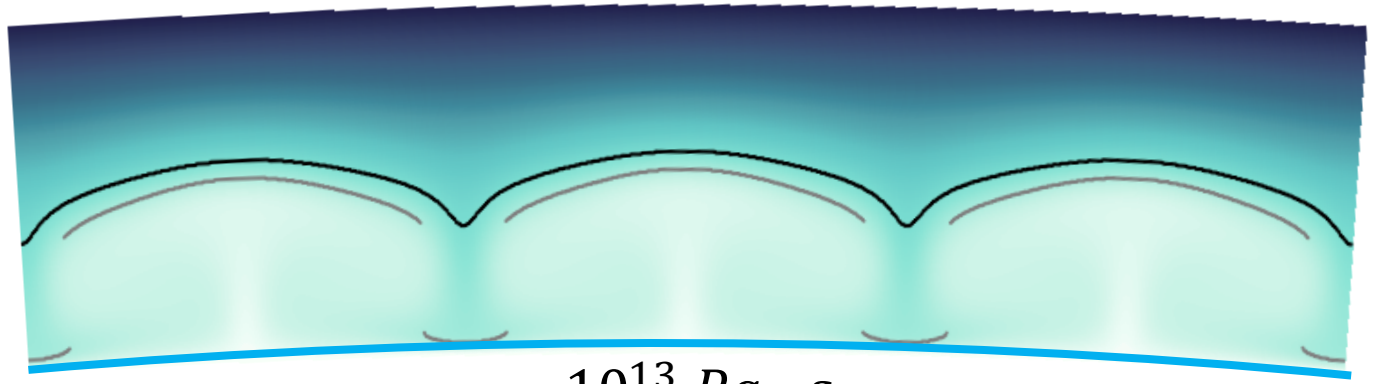




# Interface Depth Comparison

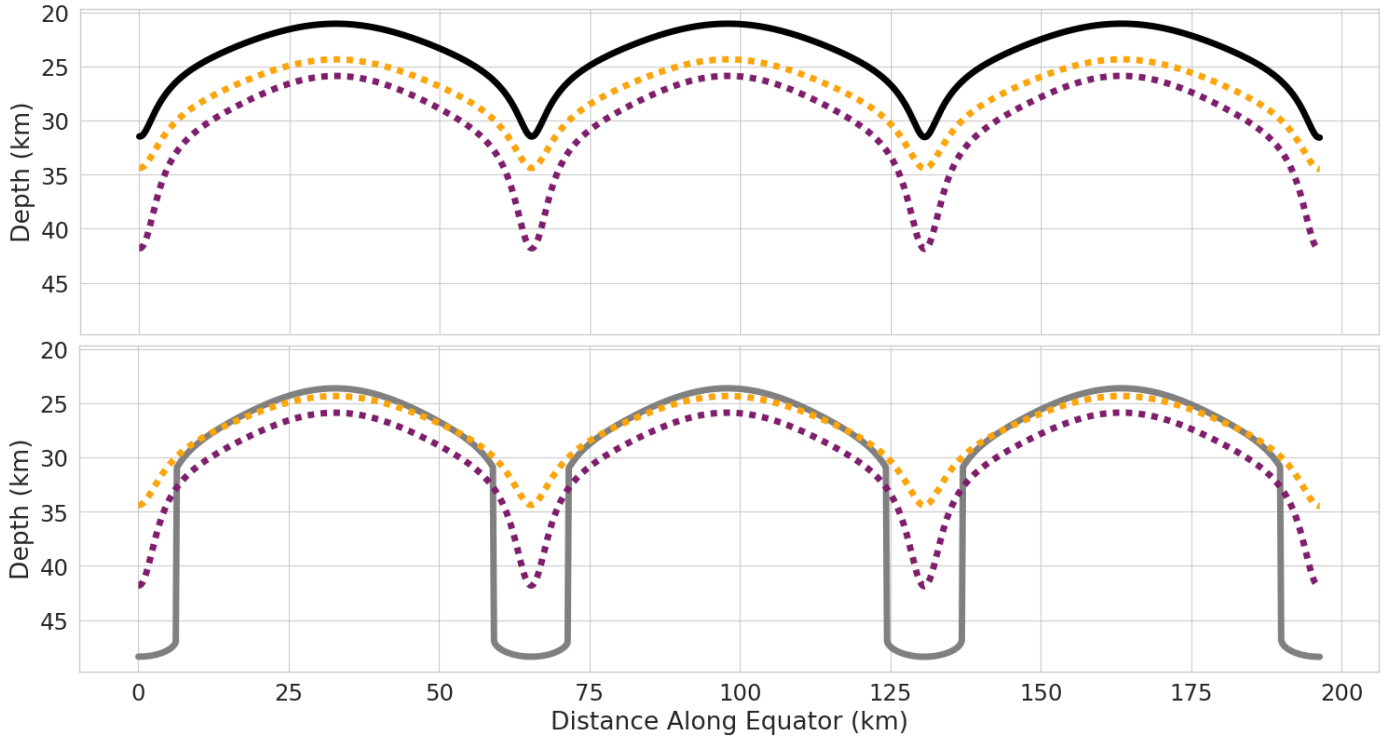
## Lateral Detectability

50 km



$10^{13}\ Pa \cdot s$




Interfaces

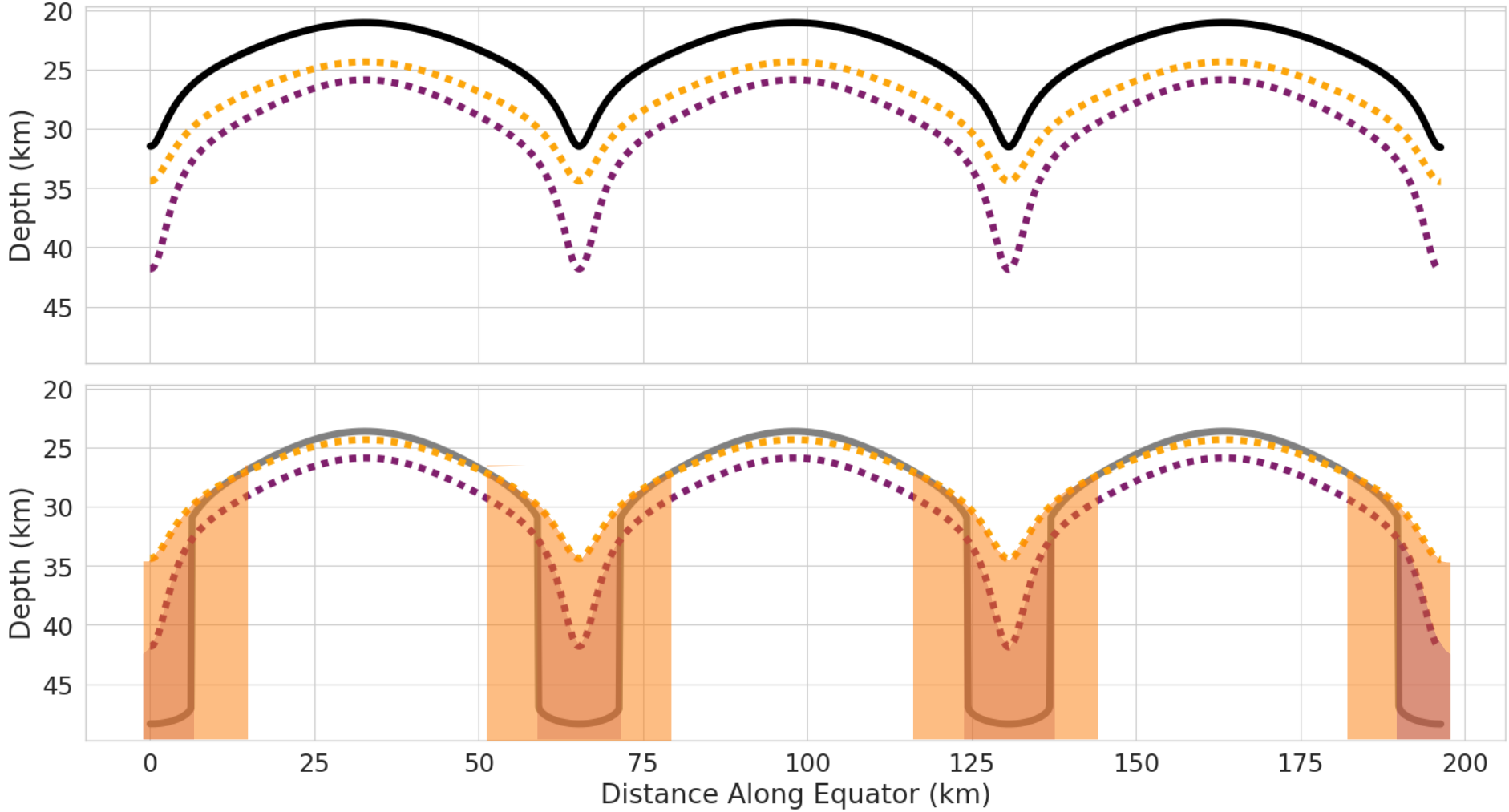
237.15 K (Multi-ion)

252.05 K ($NaCl$-$H_2O$)

High Loss (100 dB)

Low Loss (100 dB)

# Interface Depth Comparison
## Lateral Detectability



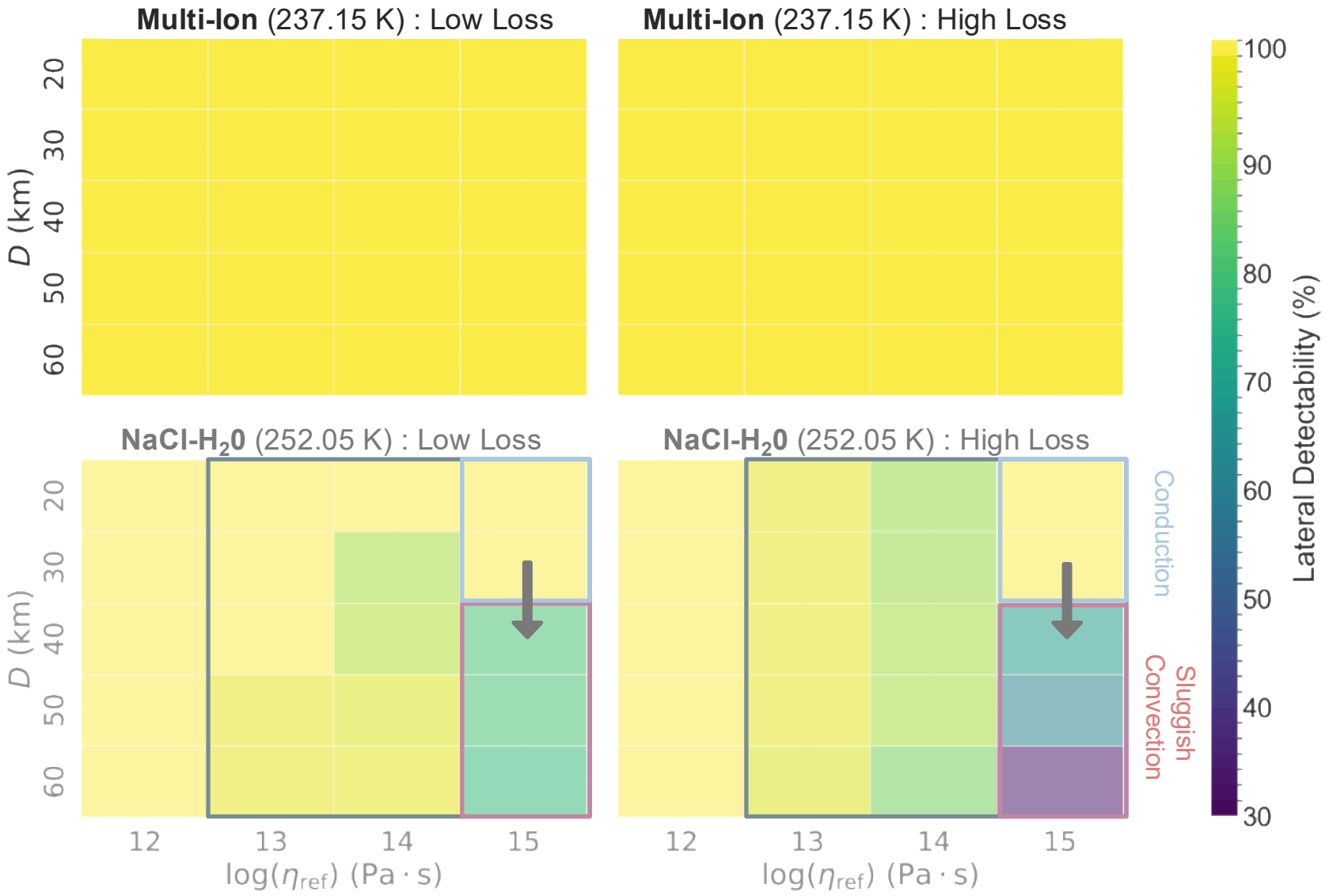

**Multi-ion eutectic:**

- Always reached before either attenuation limit
- Remains laterally detectable across the entire simulated domain

**NaCl-$H_2O$ eutectic:**

- **not detected** given the low loss attenuation model across 19.4% of simulation domain …
- **Not detected** across 40% of simulated domain with high loss model

# Detectability Matrix
## Lateral Detectability




- Across all simulations, lateral detectability **never drops** below **30%**
- **Multi-Ion** eutectic remains 100% laterally detectable **regardless of attenuation model**
- **NaCl-$H_20$** eutectic lateral detectability decreases at high viscosities ($> 10^{12}\ Pa \cdot s$)
- **Significant drop in detectability** after convective regime is reached ($Ra_{crit} < Ra$)

# Detectability Drop from Convection

## Lateral Detectability

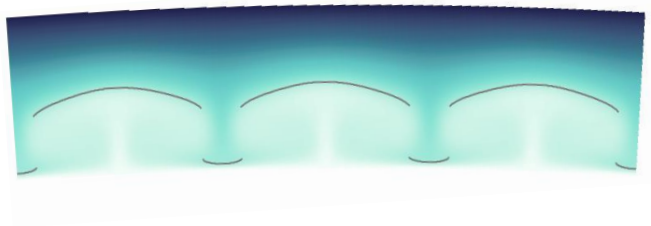
Conductive

30 km, $10^{15}\ Pa \cdot s$



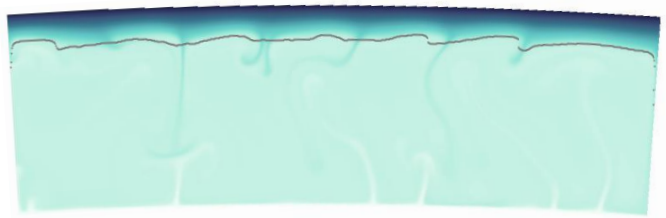
Sluggishly Convective

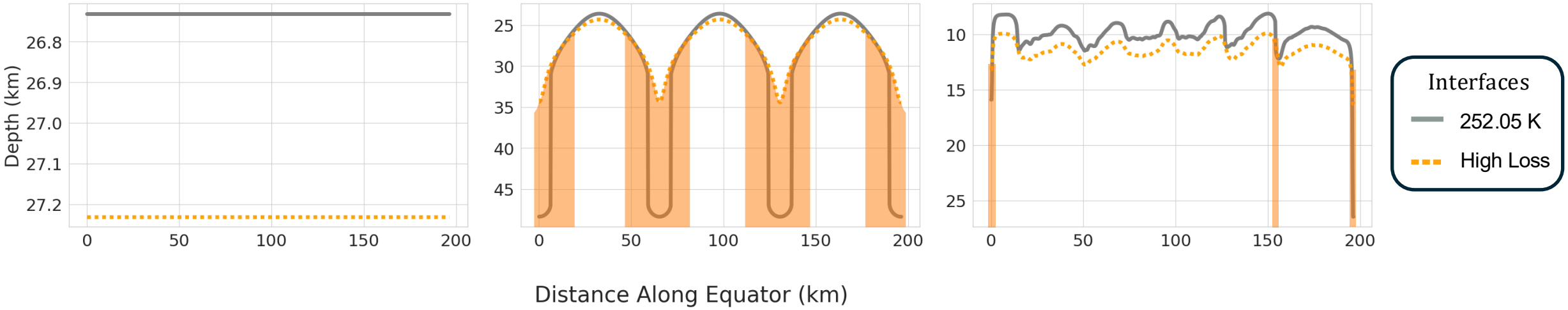
50 km, $10^{15}\ Pa \cdot s$



Vigorously Convective

60 km, $10^{13}\ Pa \cdot s$






100% Laterally Detectable

**34.1% Laterally Detectable**

98.5% Laterally Detectable

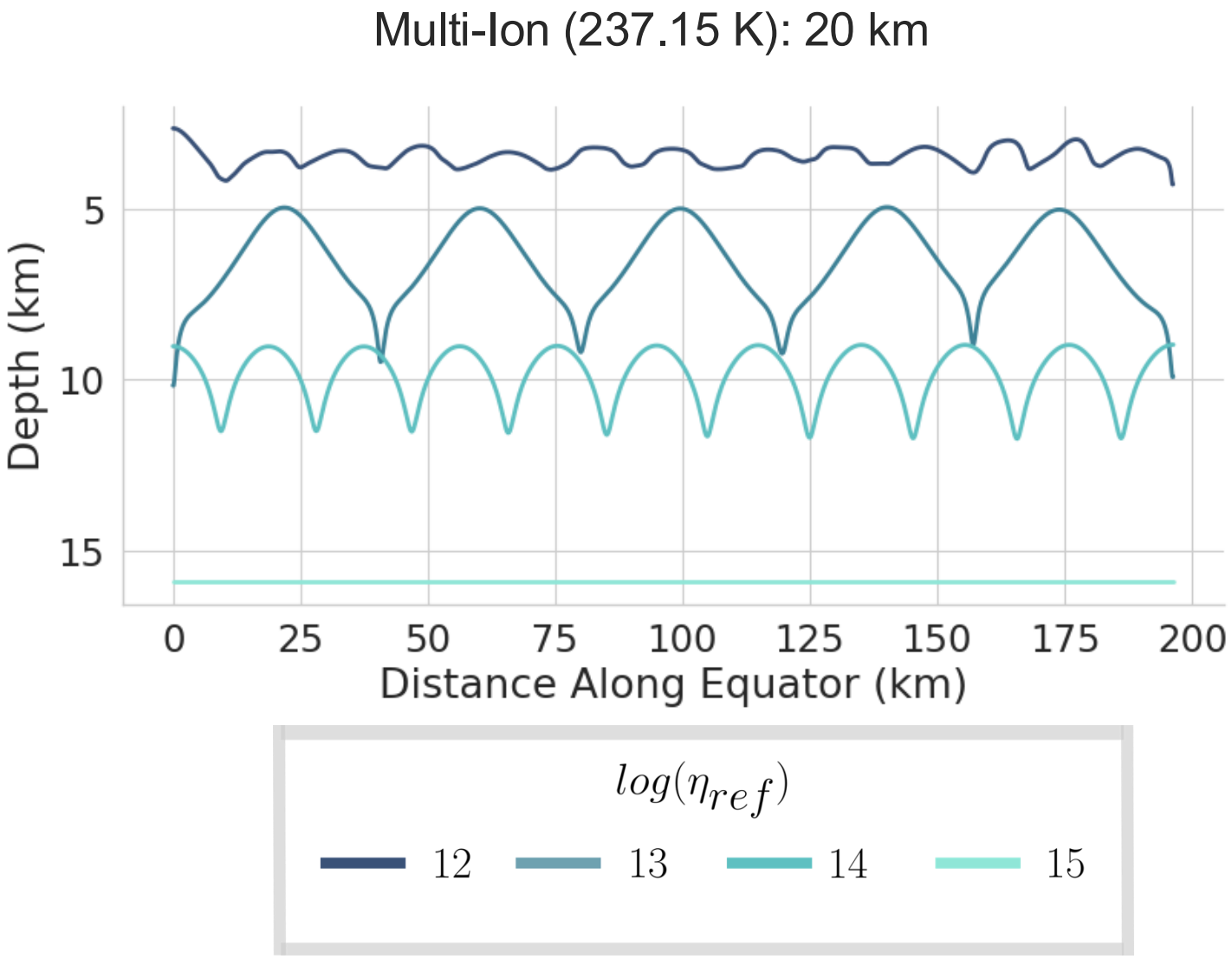
# Scientific Questions

## 1. How detectable are these eutectic interfaces?

**Likely, but decreased local detection if sluggishly convective**

## 2. Can one use properties of eutectic interfaces to geophysically characterize Europa's ice shell?

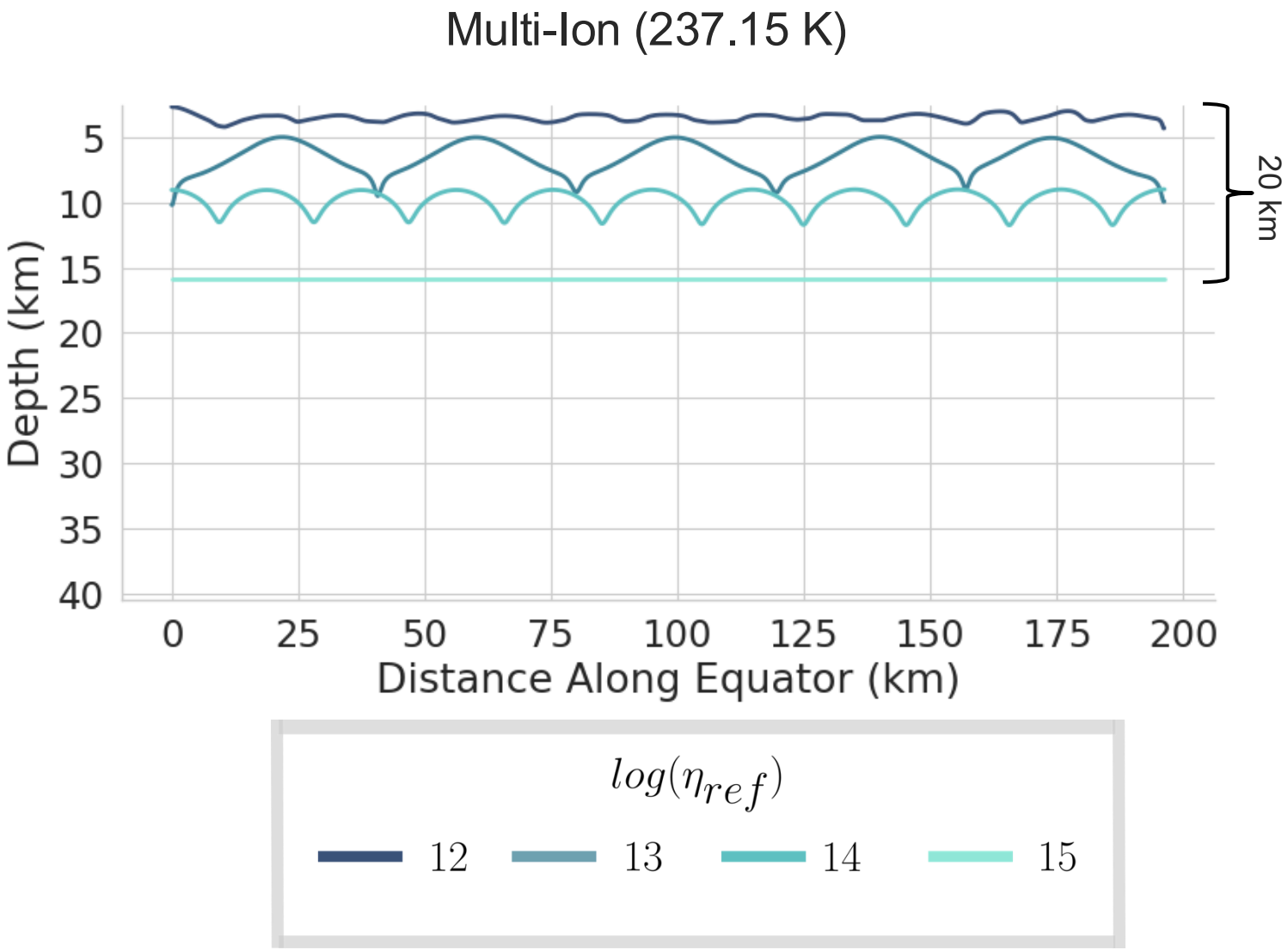
# Eutectic Interface Collection
## Geophysical Scaling Laws




For a given eutectic interface …

- There is a positive relationship between **eutectic depth** and ice shell viscosity
- And with ice shell thickness …

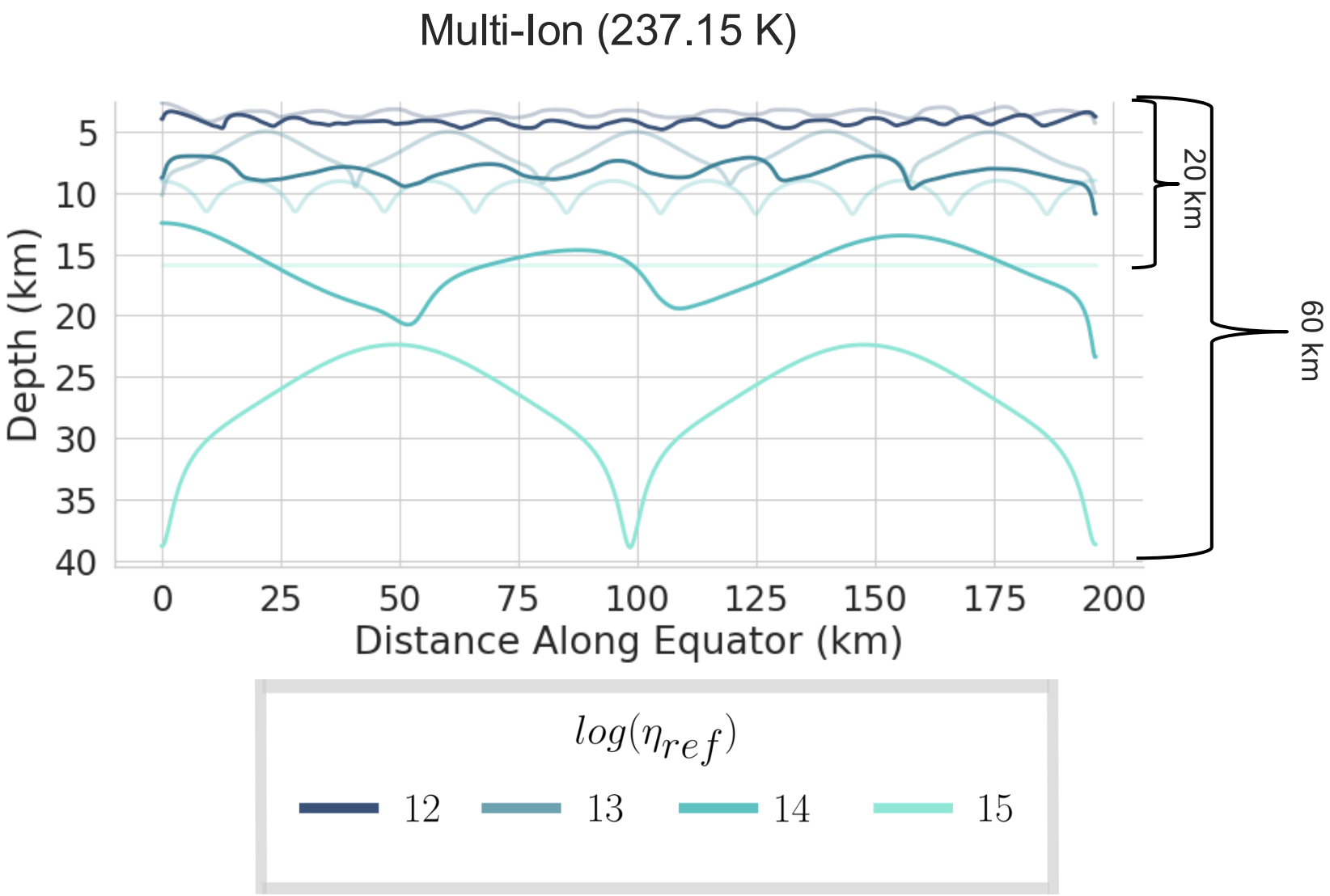
# Eutectic Interface Collection
## Geophysical Scaling Laws




For a given eutectic interface …

- There is a positive relationship between **eutectic depth** and ice shell viscosity
- And with ice shell thickness …

# Eutectic Interface Collection
## Geophysical Scaling Laws




For a given eutectic interface …

- There is a positive relationship between **eutectic depth** and ice shell viscosity
- And with ice shell thickness there is also a positive relationship with **eutectic depth**
- Is it possible to develop some form of mathematical scaling law?

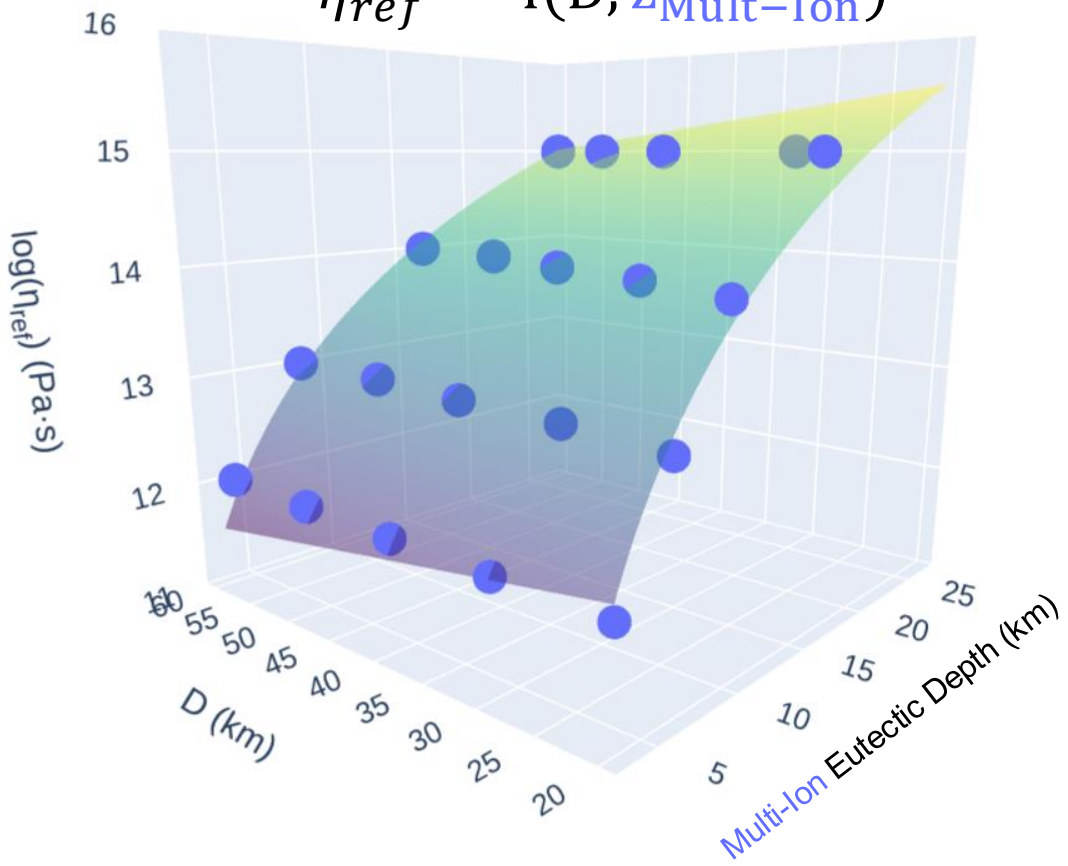
# Fits & Solutions: Log Bend Example

## Geophysical Scaling Laws

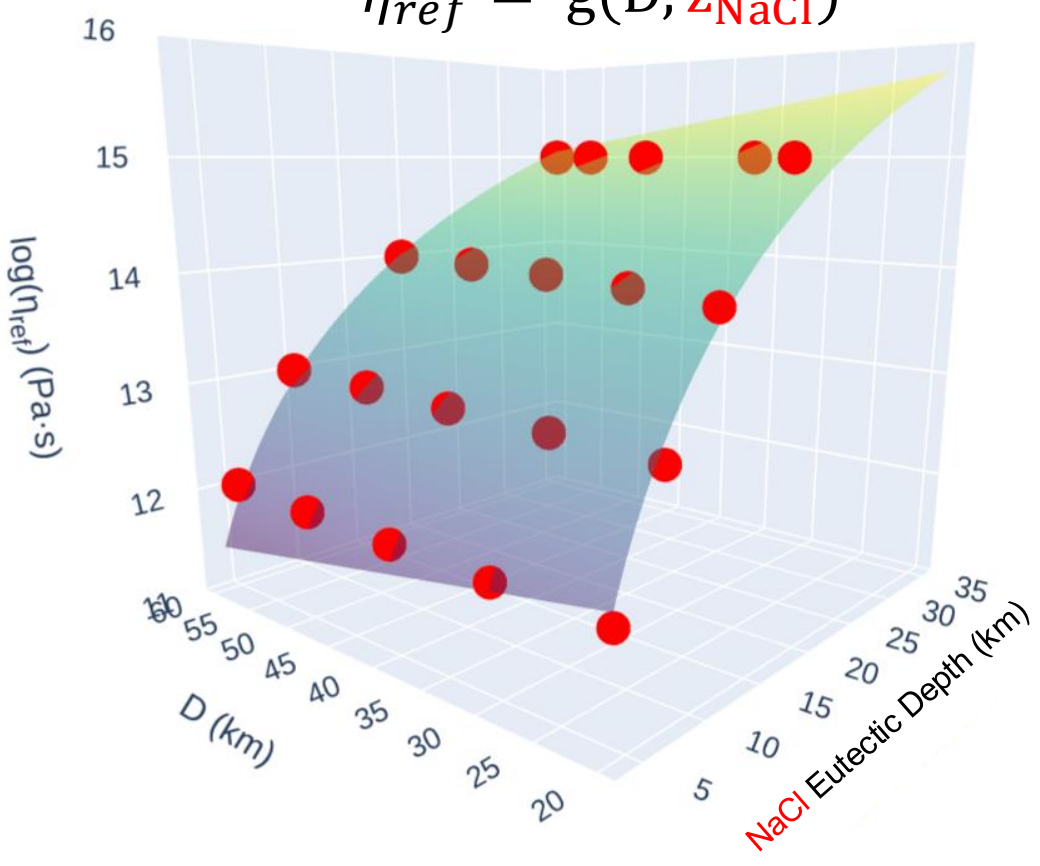
$\bar{z}$ = Mean Eutectic depth

**Log Bend:** $\eta_{ref} = a + b\bar{z} + cD + d\boldsymbol{log}(\bar{z})$

$\eta_{ref} = f(D, \bar{z}_{Mult-Ion})$




$\eta_{ref} = g(D, \bar{z}_{NaCl})$

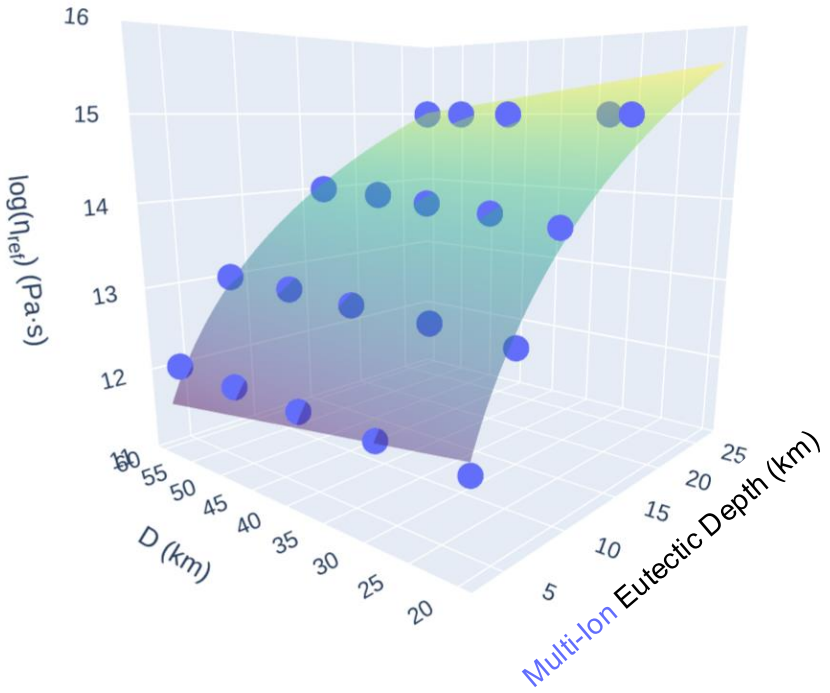


# Fits & Solutions: Log Bend Example
## Geophysical Scaling Laws

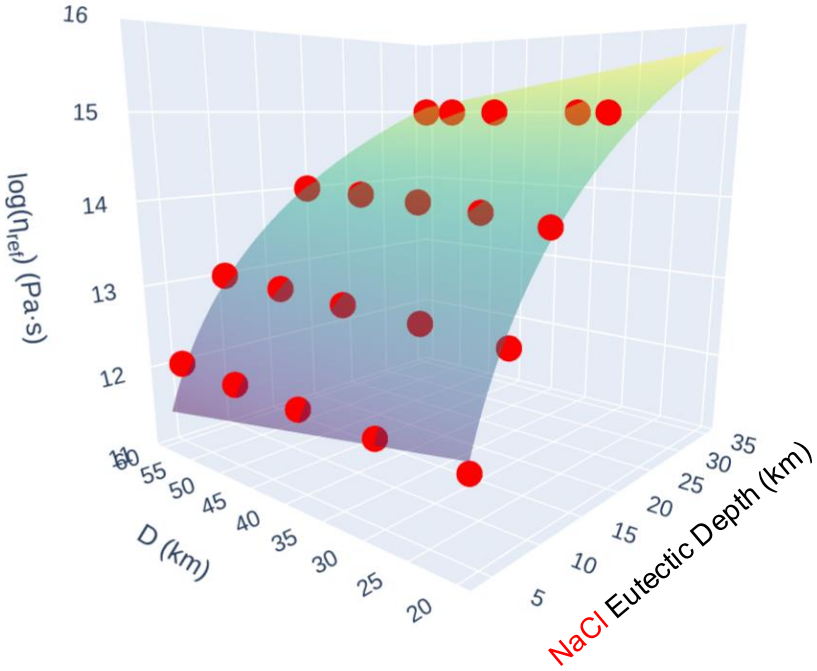
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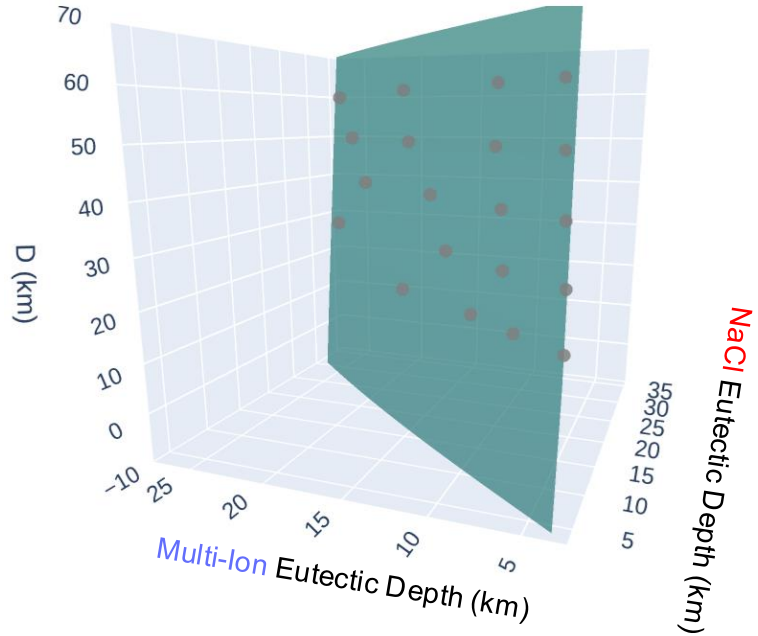
$\eta_{ref} = f(D, \bar{z}_{Mult-Ion})$




$\eta_{ref} = g(D, \bar{z}_{NaCl})$



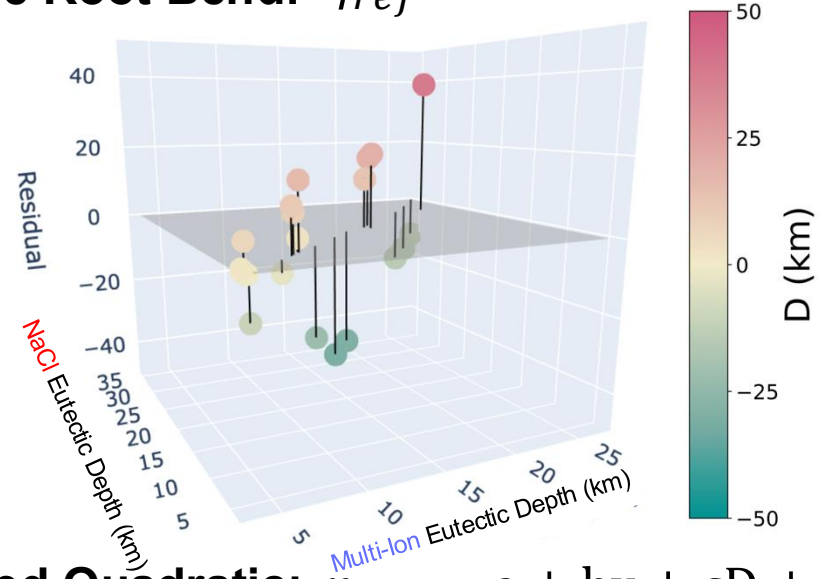

$D = h(\bar{z}_{Mult-Ion}, \bar{z}_{NaCl})$



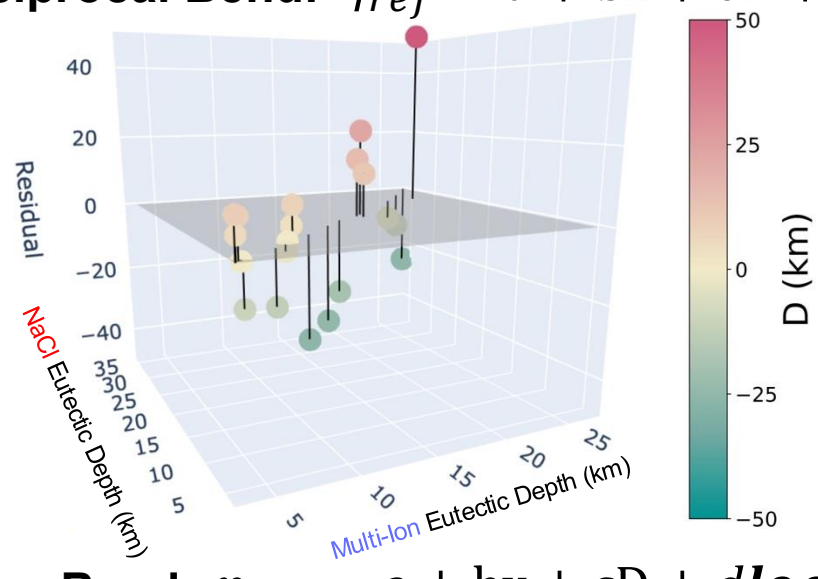
# Solution Surfaces

## Geophysical Scaling Laws

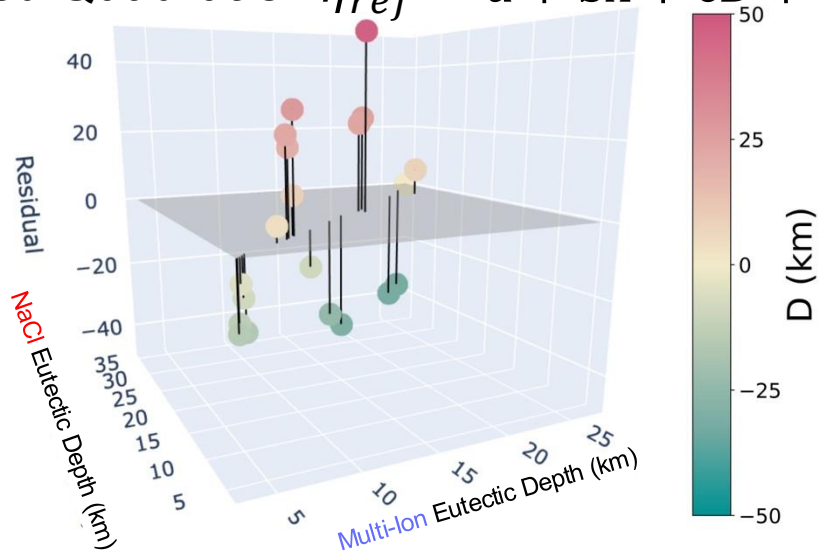
**Square Root Bend:** $\eta_{ref} = \mathrm{a} + \mathrm{bx} + \mathrm{cD} + d\bar{z}^{1/2}$



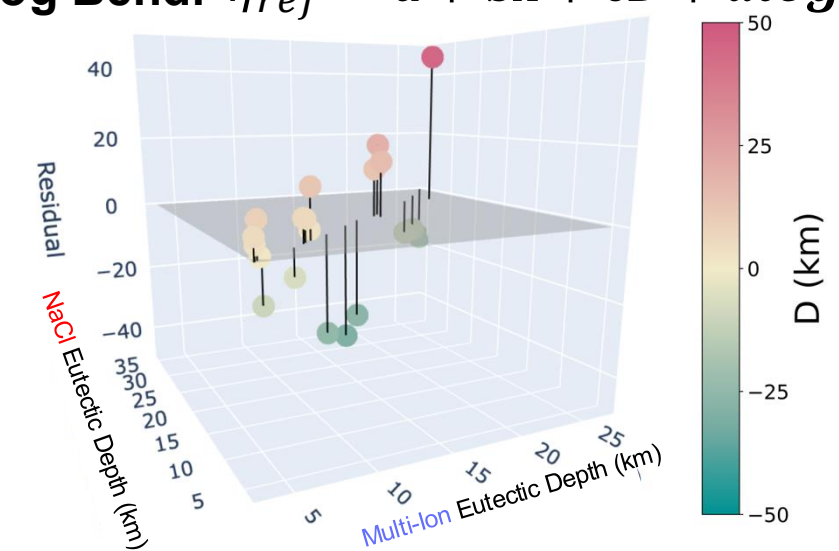

**Reciprocal Bend:** $\eta_{ref} = \mathrm{a} + \mathrm{bx} + \mathrm{cD} + d\bar{z}^{-1}$



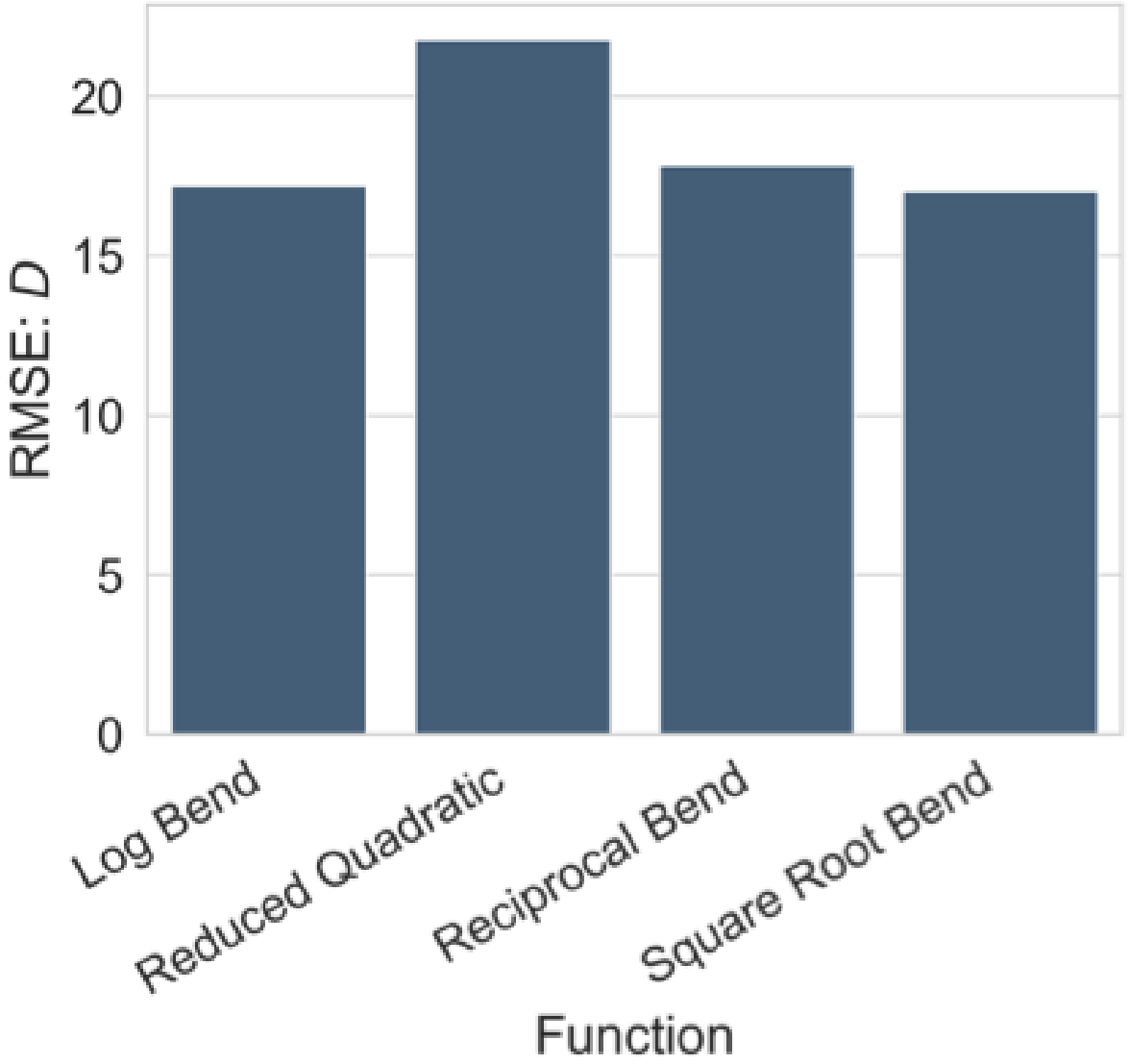

**Reduced Quadratic:** $\eta_{ref} = \mathrm{a} + \mathrm{bx} + \mathrm{cD} + d\bar{z}^{2}$

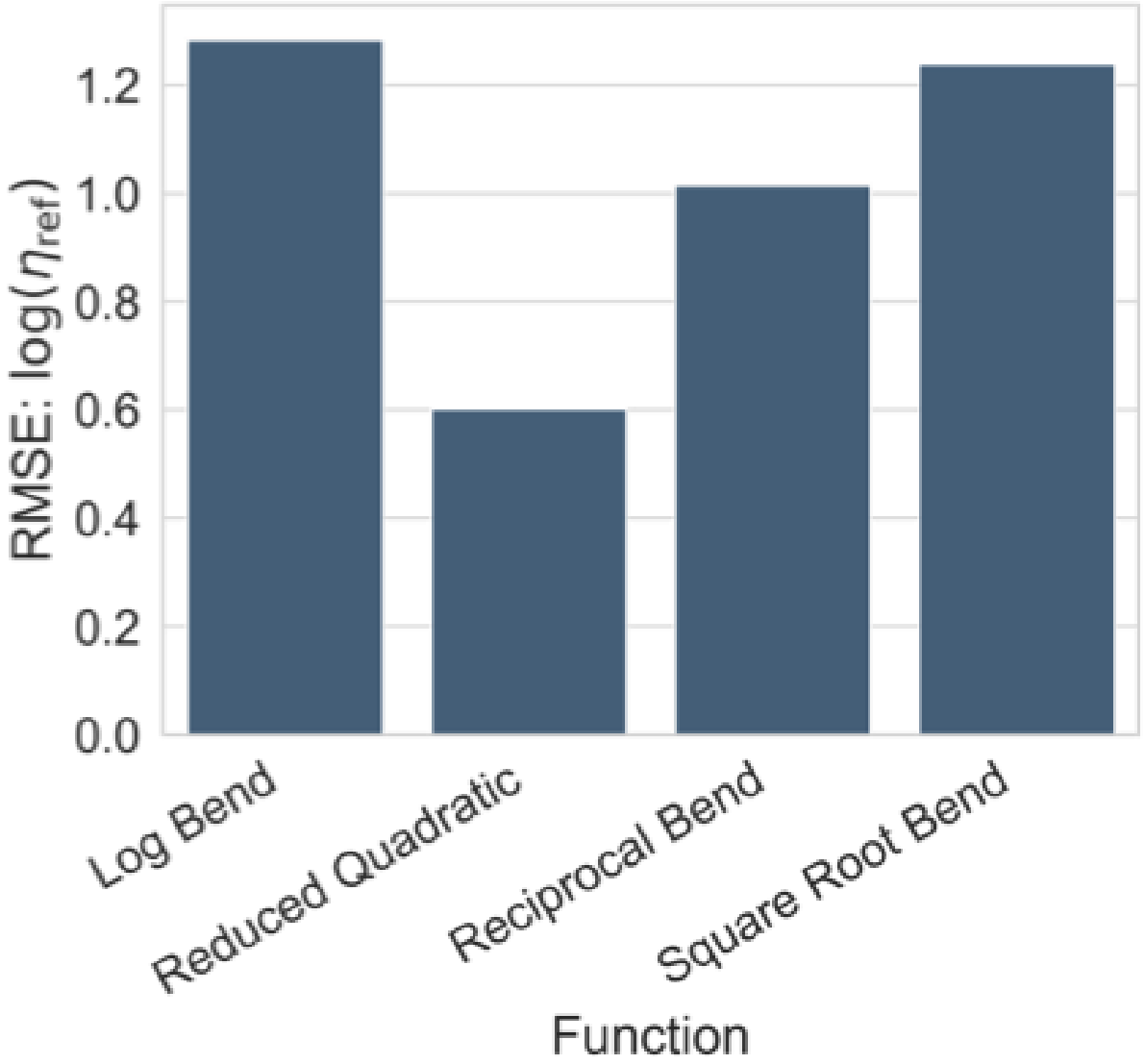



**Log Bend:** $\eta_{ref} = \mathrm{a} + \mathrm{bx} + \mathrm{cD} + d\boldsymbol{log}(\bar{z})$



# RMSE Comparison
## Geophysical Scaling Laws



# Scientific Questions

## 1. How detectable are these eutectic interfaces?

**Yes, with decreased local detection if sluggishly convective**

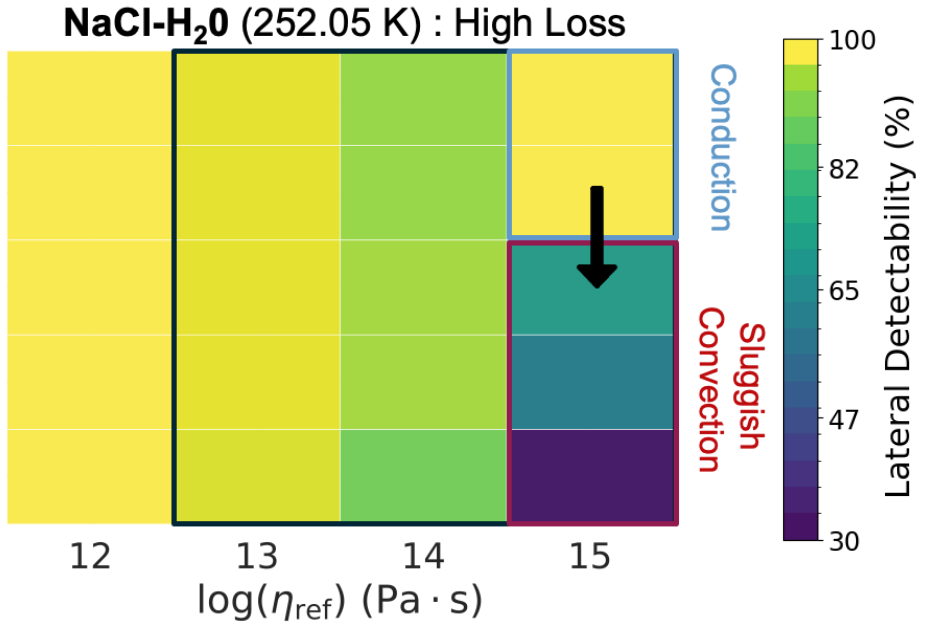
## 2. Can one use properties of eutectic interfaces to geophysically characterize Europa's ice shell?

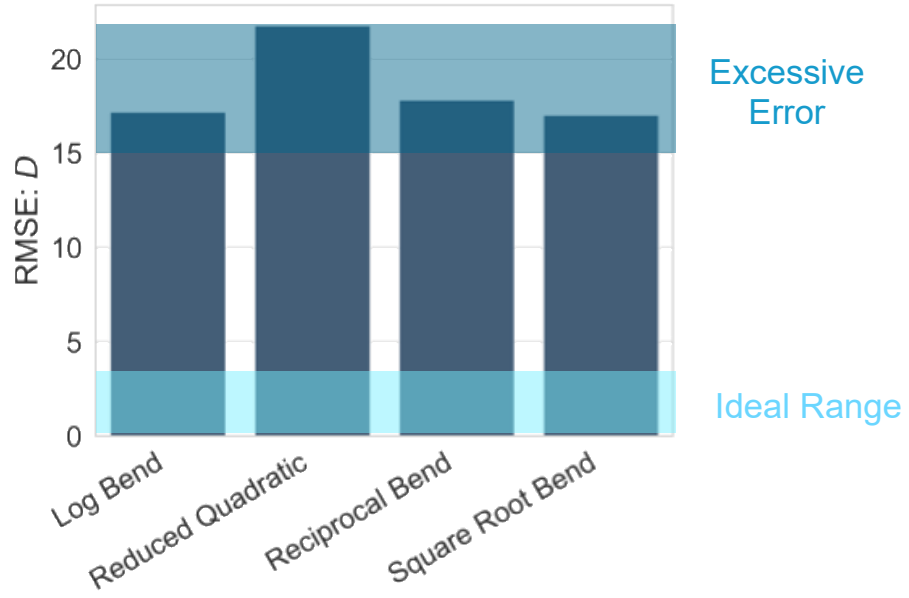
**Potentially. Current methods require more data to reduce errors**

# Conclusion

- Eutectic **interface detectability** remains **possible across our entire parameter study**, with a notable **decrease** laterally for **sluggishly convective** simulations of up to **34.1%**.

- We **identified scaling relationships** between **mean eutectic depth** and **shell thickness & reference viscosity**, however our limited dataset shows a weak relation

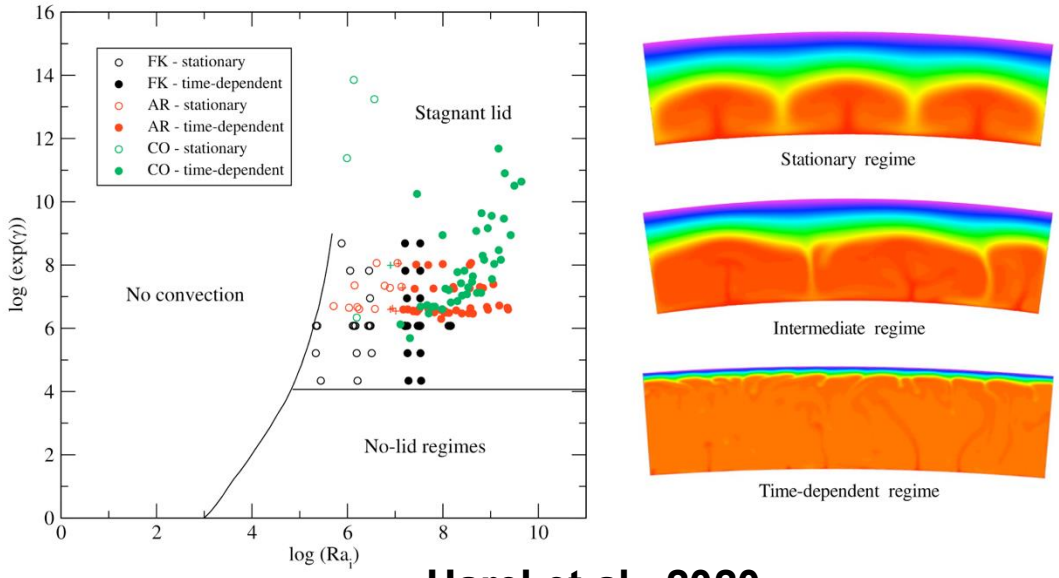


# Future Work

## Decreased Error

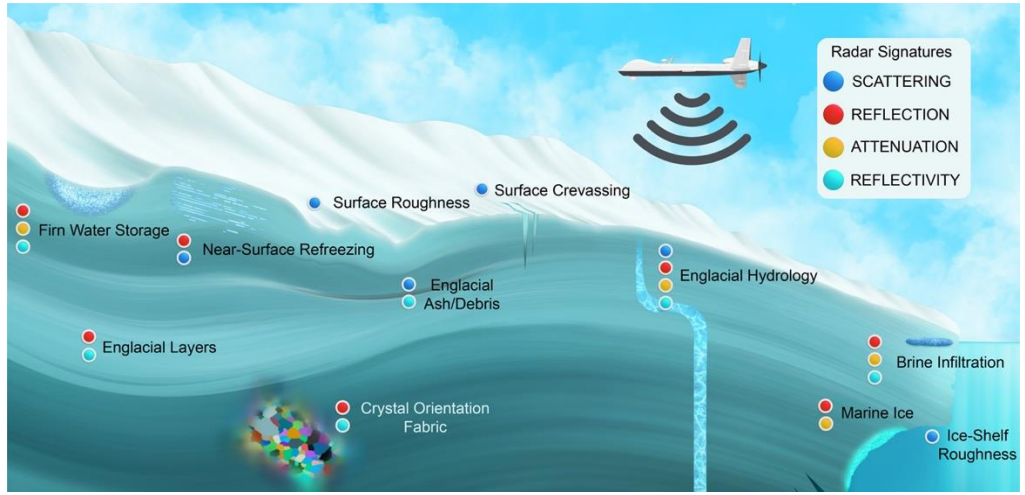
- **Grouping scaling laws** according to **convective regimes**
- Perform a higher **resolution parameter study**




**Harel et al., 2020**

## More realistic results

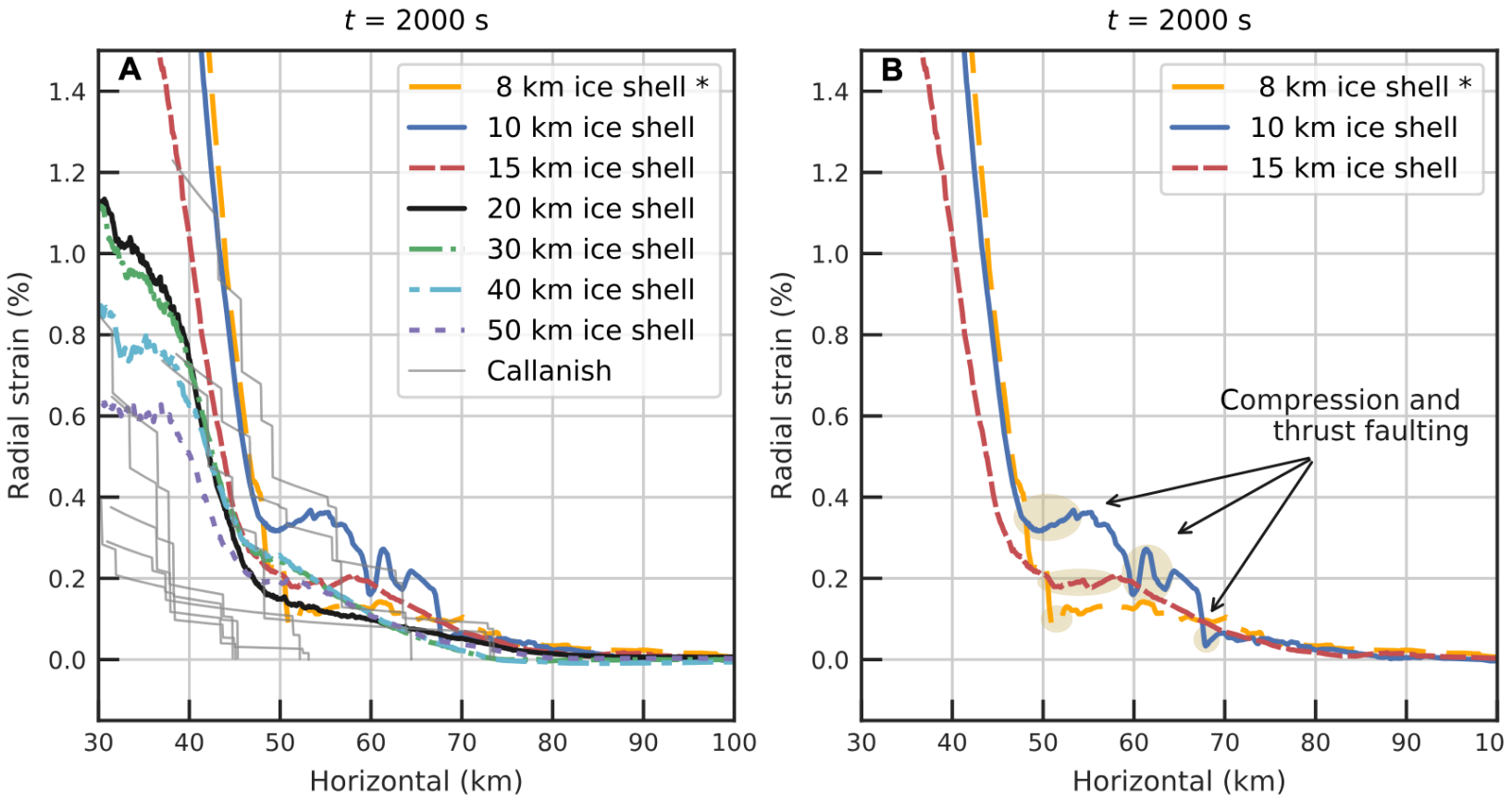
- Model **additional reductions** to radar **sounding losses** (besides attenuation)




**Schroeder, 2022**

# Backup Slides

# Why Minimum 20 km Ice shell?
## Simulations

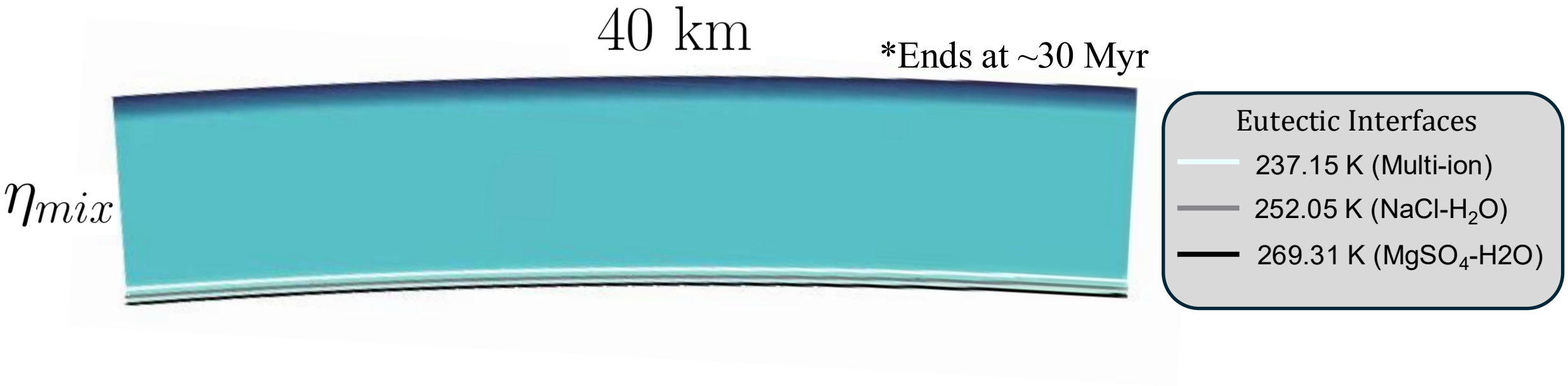



**Wakita et al., 2024**

iSale-2D simulations of multi-ring basins

- Callanish: ~100 km diameter
- Both basins show only extensional features
- Compressional features only occur if D < 20 km

# Why No $MgSO_4$-$H_2O$ Eutectic?
## Simulations




- $MgSO_4$-$H_2O$ eutectic is too close to the ice-ocean interface

# GAIA Input Parameters (Baseline)
## Simulations

Table 1: Costants Used Across Thermal Evolution Models

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| [a]Density | $\rho_{ref}$ | 920 | $kg \cdot m^{-3}$ |
| [b]Gravitational Acceleration | $g$ | 1.315 | $m \cdot s^{-2}$ |
| [c]Surface Temperature | $T_{surf}$ | 110 | $K$ |
| [b]Specific Heat Capacity | $c_p$ | 2090 | $J \cdot kg^{-1} \cdot K^{-1}$ |

[a] Hobbs (1974)
[b] Schubert, Sohl, and Hussmann (2009)
[c] Ojakangas (1989)

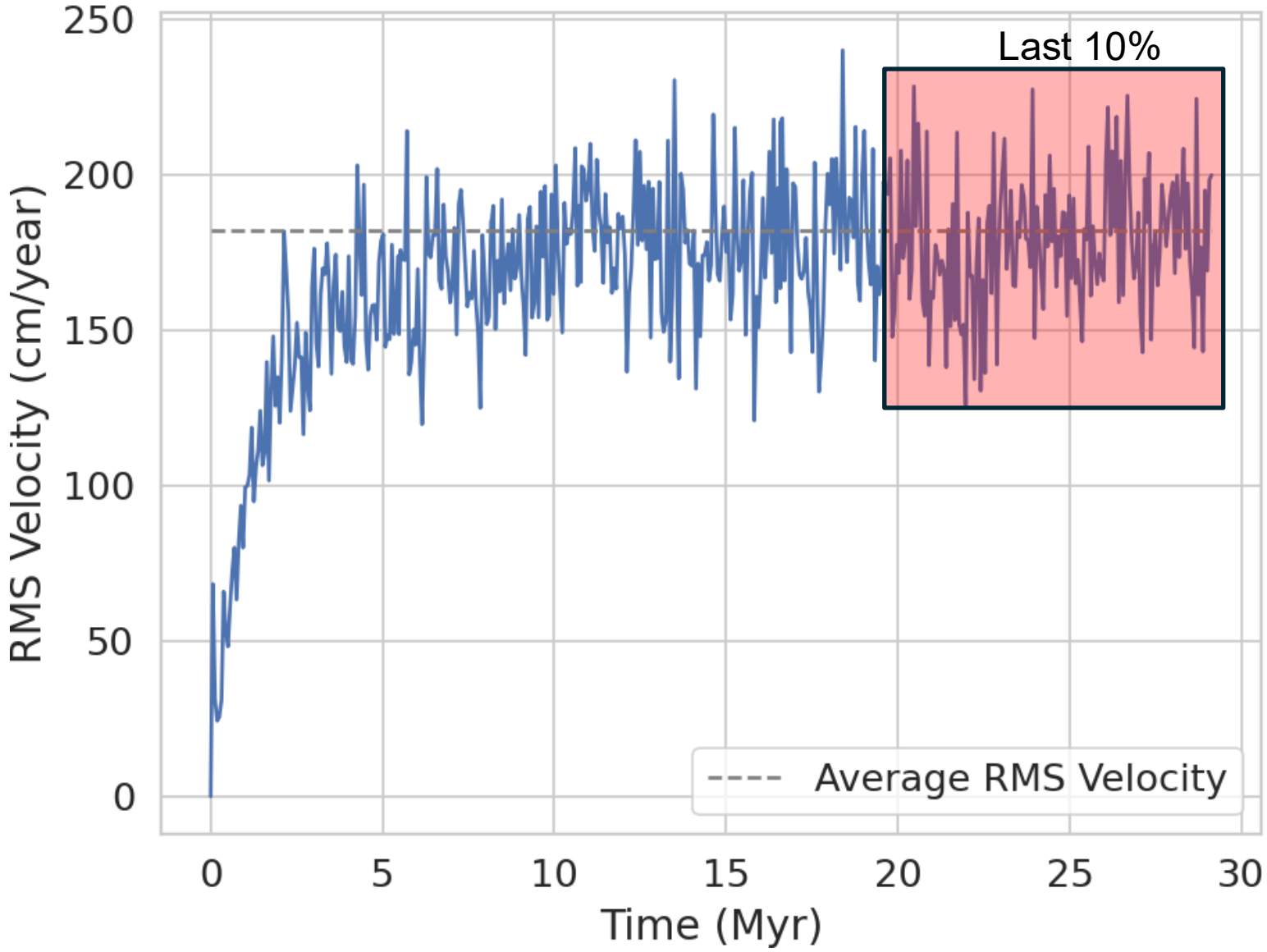
Table 2: Ice Shell Thickness Varying Parameters for Thermal Evolution Model

| [a]$D$ (km) | [a]$T_b$ (K) | [b]$\alpha_{ref}$ ($K^{-1}$) | [c]$k_{ref}$ (W $m^{-1}$ $K^{-1}$) | $\kappa_{ref}$ ($m^2$ $s^{-1}$) |
|---|---|---|---|---|
| 20 | 271.2 | $1.544 \cdot 10^{-4}$ | 2.400 | $1.248 \cdot 10^{-6}$ |
| 30 | 270.2 | $1.518 \cdot 10^{-4}$ | 2.409 | $1.252 \cdot 10^{-6}$ |
| 40 | 269.2 | $1.492 \cdot 10^{-4}$ | 2.418 | $1.257 \cdot 10^{-6}$ |
| 50 | 268.1 | $1.465 \cdot 10^{-4}$ | 2.428 | $1.262 \cdot 10^{-6}$ |
| 60 | 267.0 | $1.438 \cdot 10^{-4}$ | 2.439 | $1.268 \cdot 10^{-6}$ |

[a] Schubert, Sohl, and Hussmann (2009)
[b] Feistel and Wagner (2006)
[c] Petrenko and Whitworth (1999)

# Reaching a Steady State

## Simulations




- Reach Steady state
  - Run for at least 20 Myr
- All analysis done within an averaging window of the last 10% of timesteps
- This avoids biased analysis (e.g. just after convective cell collapse)

# Fractional Polynomials
## Geophysical Scaling Laws

Assuming x is mean eutectic interface depth, we have:

$$\eta_{ref} = \mathrm{f}(\mathrm{x}_{237}, \mathrm{D})$$
$$\eta_{ref} = \mathrm{g}(\mathrm{x}_{252}, \mathrm{D})$$

Solving for D:

$$\mathrm{D} = \mathrm{h}(\mathrm{x}_{237}, \mathrm{x}_{252})$$

Why these functions?

- Low coefficient number prevents overfitting for small datasets
- Only one solution compared to full quadradic or cubic

**Reduced Quadratic**

1. $\eta_{ref} = \mathrm{a} + \mathrm{bx} + \mathrm{cD} + \mathrm{dx}^2$

**Square Root Bend**

2. $\eta_{ref} = \mathrm{a} + \mathrm{bx} + \mathrm{cD} + \mathrm{dx}^{1/2}$

**Log Bend**

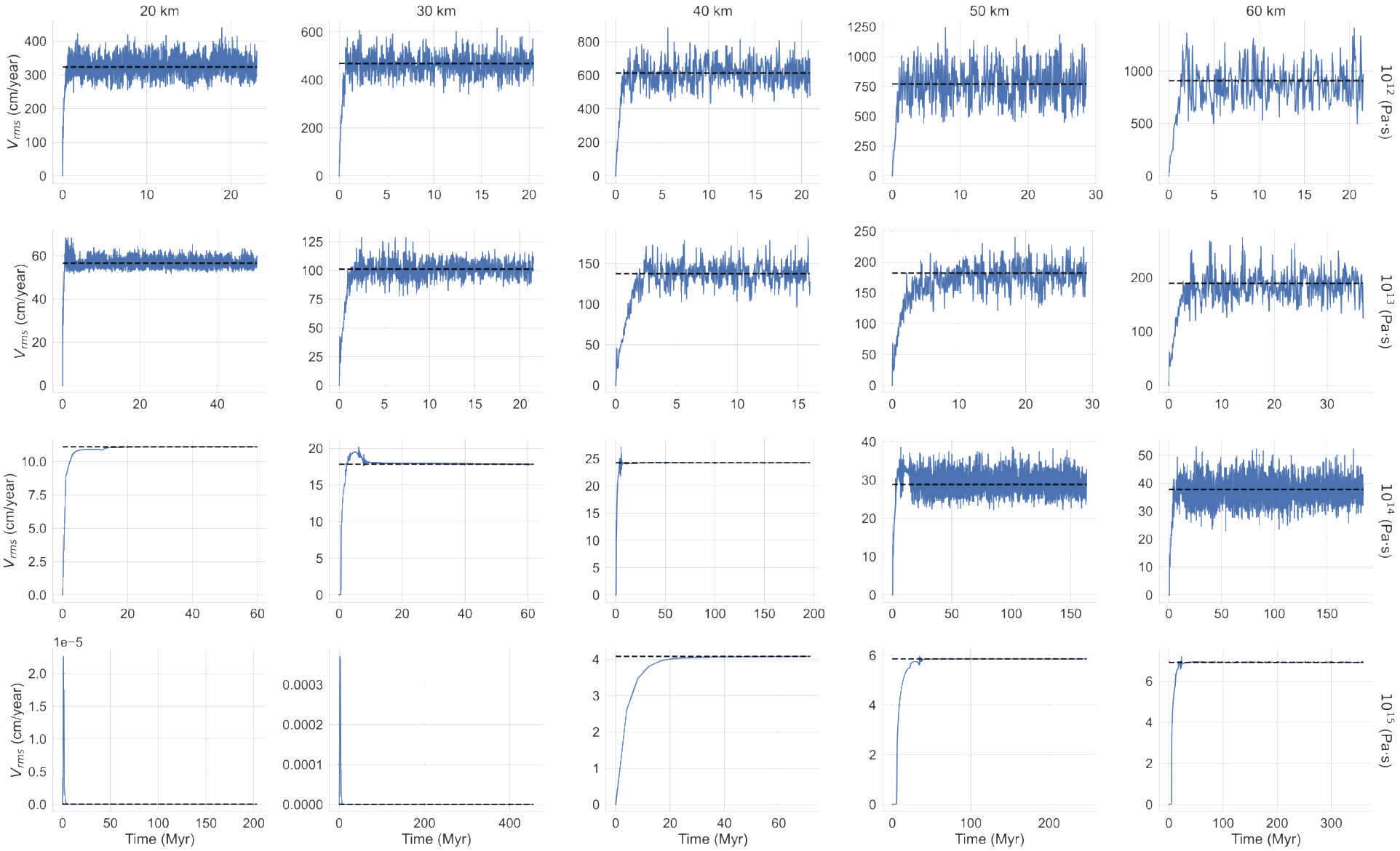
3. $\eta_{ref} = \mathrm{a} + \mathrm{bx} + \mathrm{cD} + \mathrm{dlog(x)}$

**Reciprocal Bend**

4. $\eta_{ref} = \mathrm{a} + \mathrm{bx} + \mathrm{cD} + \mathrm{dy}^{-1}$

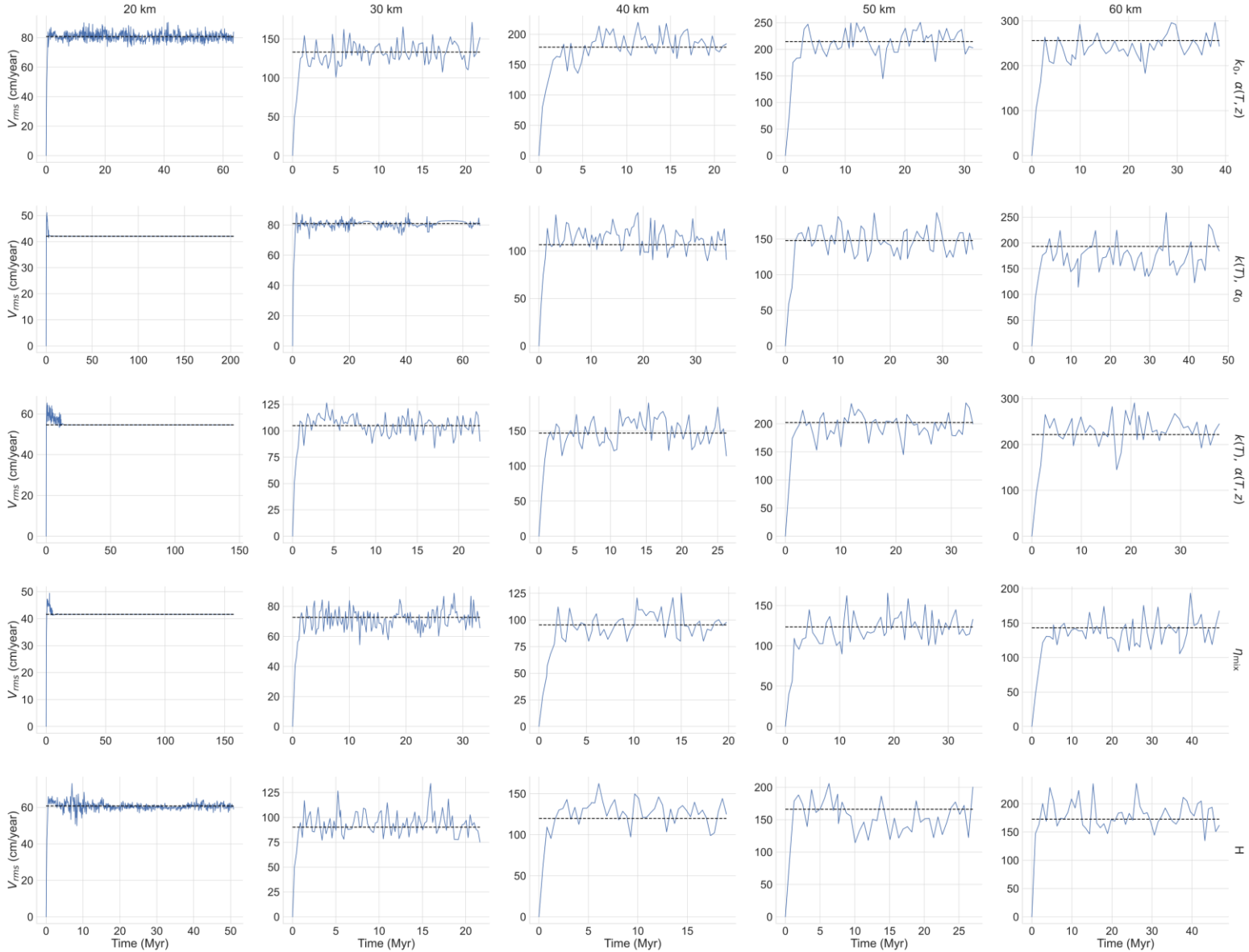
# Baseline RMS Velocity

## Simulations



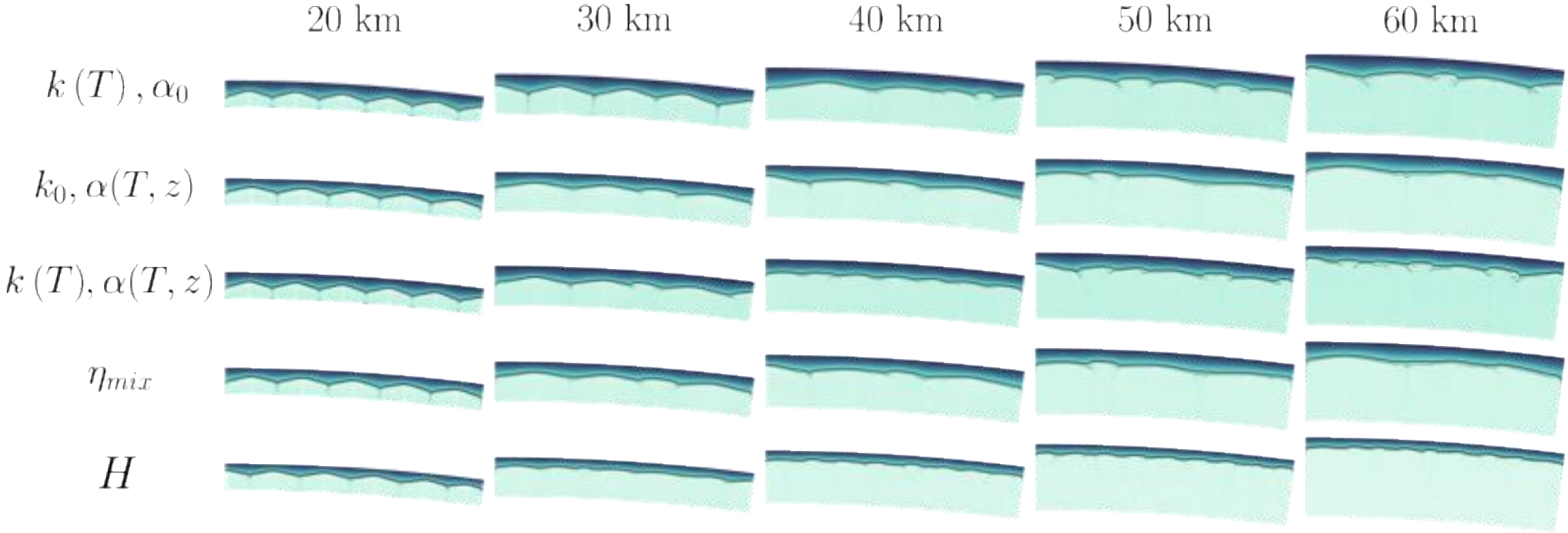
# Non-Baseline RMS Velocity

## Simulations



# Non-Baseline Temperature Fields

## Simulations




Additional Parametric Variation

$k(T)$: Variable thermal conductivity

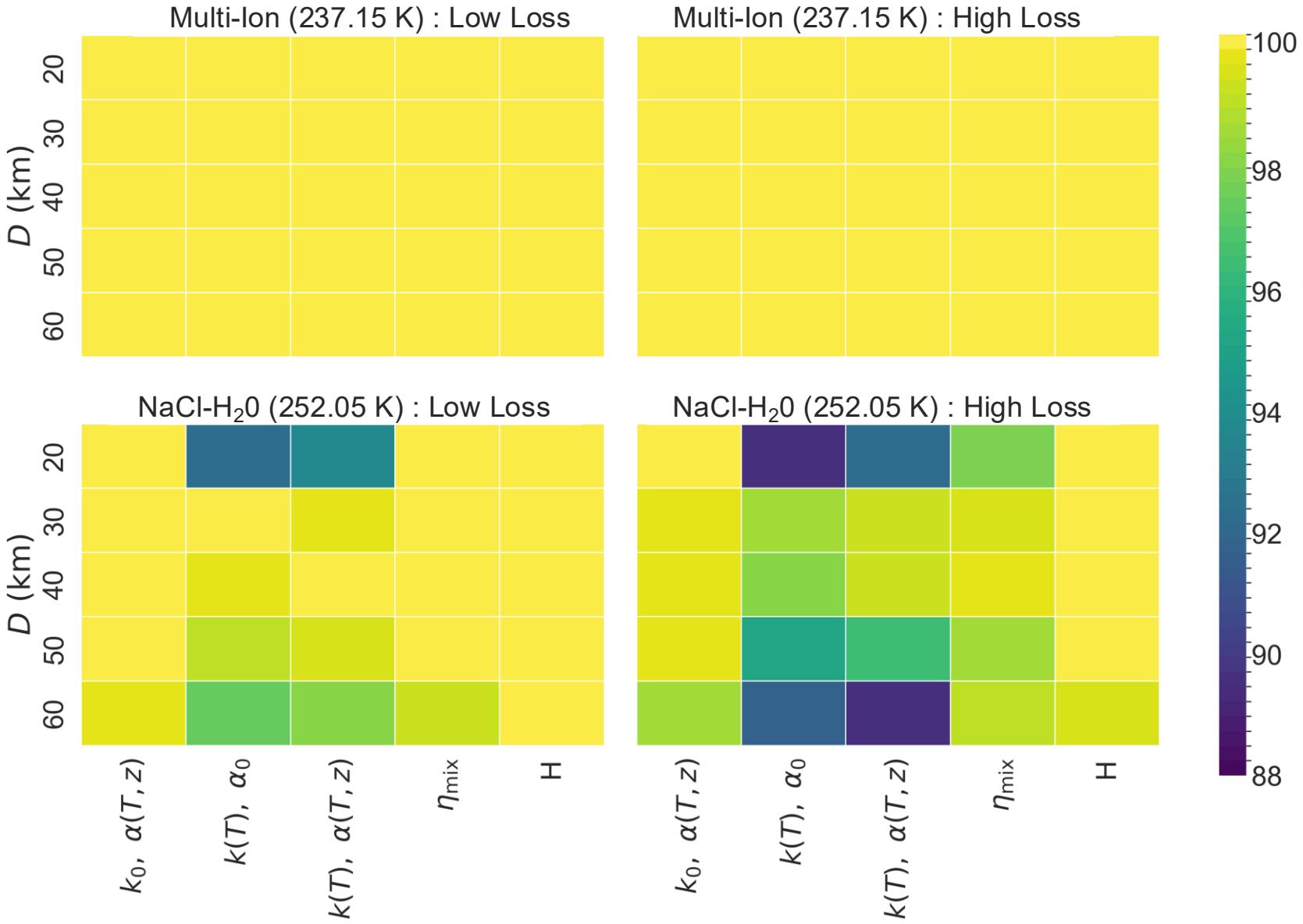
$\alpha(T,\mathrm{z})$: Variable thermal expansivity

$\eta_{mix}$: Composite Rheology ($\eta_{diff}, \eta_{disl}, \eta_{gbs+bs}$)

H: Tidal Heating

# Non-Baseline Detectability Matrix

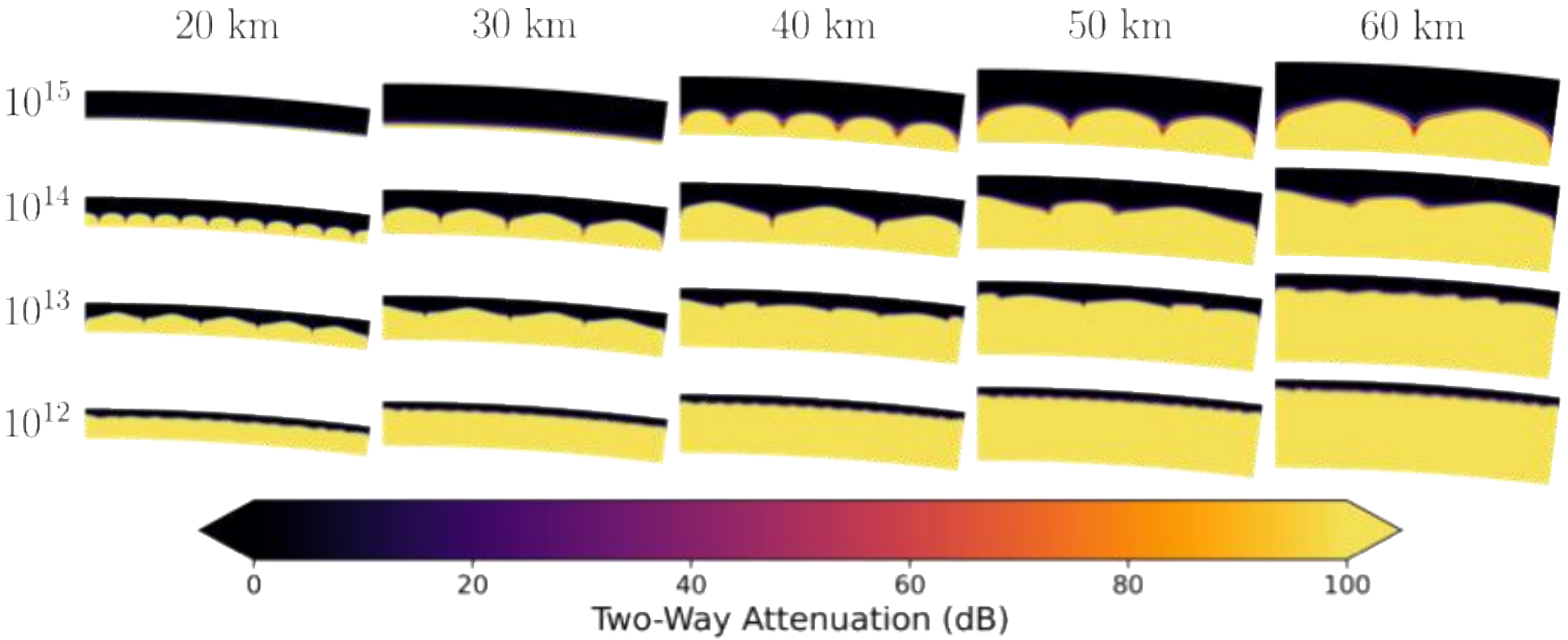
## Lateral Detectability




- All simulations with fixed $\eta_{ref}$ of $\mathbf{10^{13}}\ \boldsymbol{Pa \cdot s}$
- Same trend for colder (shallower) multi-ion brine eutectic
- For the NaCl-$H_2O$, drop in lateral detectability at very small and large ice shell thicknesses
- Possible due to random plume destabilization
- Implies weak relation between ice shell thickness and lateral detectability

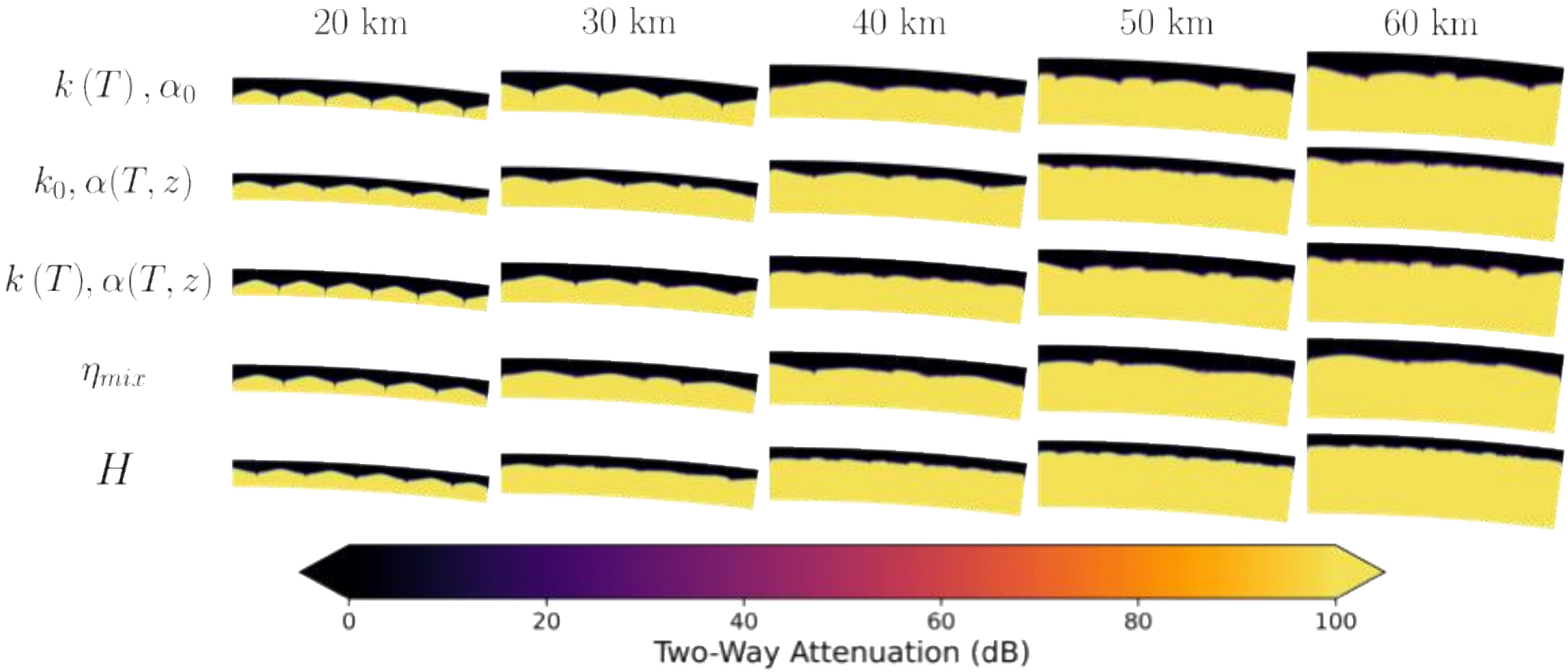
# Baseline Attenuation Field

## Lateral Detectability



# Non-Baseline Attenuation Field

## Lateral Detectability



# Tidal Heating Scaling
# Non-Baseline: Tidal Heating

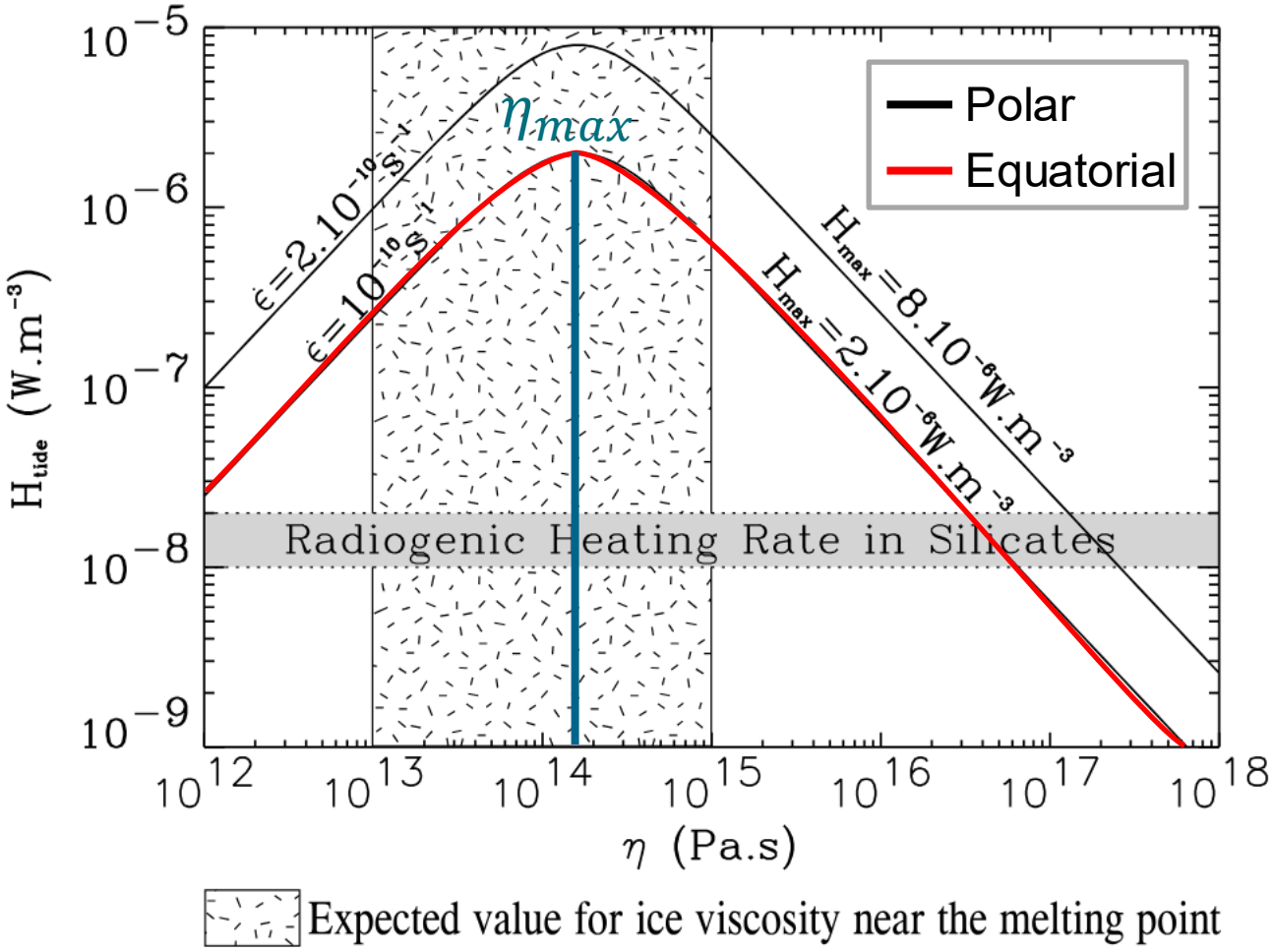
$$H = \frac{2H_{max}}{\frac{\eta}{\eta_{max}} + \frac{\eta_{max}}{\eta}}$$

H: Actual Tidal Heating ($W \cdot m^{-3}$)

$H_{max}$: Tidal Heating Maximum

$\eta$: Actual Viscosity ($Pa \cdot s$)

$\eta_{max}$: Viscosity experienced at $H_{max}$




**Tobie et al., (2003)**

# Rheological Parameters
## Non-Baseline: Complex Rheology

Table 3: Rheological Parameters Per Deformation Mechanism

| Mechanism | $n$ | $m$ | $A_{cold}$ $(m^m Pa^{-n} s^{-1})$ | $E_{cold}$ (kJ mol$^{-1}$) | $A_{warm}$ $(m^m Pa^{-n} s^{-1})$ | $E_{warm}$ (kJ mol$^{-1}$) |
|---|---|---|---|---|---|---|
| Diffusion creep | 1.0 | 2.0 | $3.3 \times 10^{-10}$ | 59.4 | $3.3 \times 10^{-10}$ | 59.4 |
| Dislocation creep | 4.0 | 0.0 | $4.0 \times 10^{-19}$ | 60.0 | $6.0 \times 10^{4}$ | 180.0 |
| Grain-boundary sliding | 1.8 | 1.4 | $6.2 \times 10^{-14}$ | 49.0 | $4.8 \times 10^{15}$ | 192.0 |
| Basal slip | 2.4 | 0.0 | $2.2 \times 10^{-7}$ | 60.0 | $2.2 \times 10^{-7}$ | 60.0 |

* Values from Goldsby and Kohlstedt (2001) and Durham, Stern, and Kirby (2001). $A$ and $E$ differ above/below the transition temperatures of 258 K (dislocation creep) and 255 K (grain-boundary sliding), denoted warm/cold.

Complex rheology simulations ($\eta_{ref}$) given grain size of 0.16 mm

- Solve for grain size given diffusion with $\eta = 10^{13} Pa \cdot s$

# Diffusion
## Non-Baseline: Complex Rheology

Deformation mechanisms determine how material flows in the shell

- Modifies viscosity per cell, per time-step

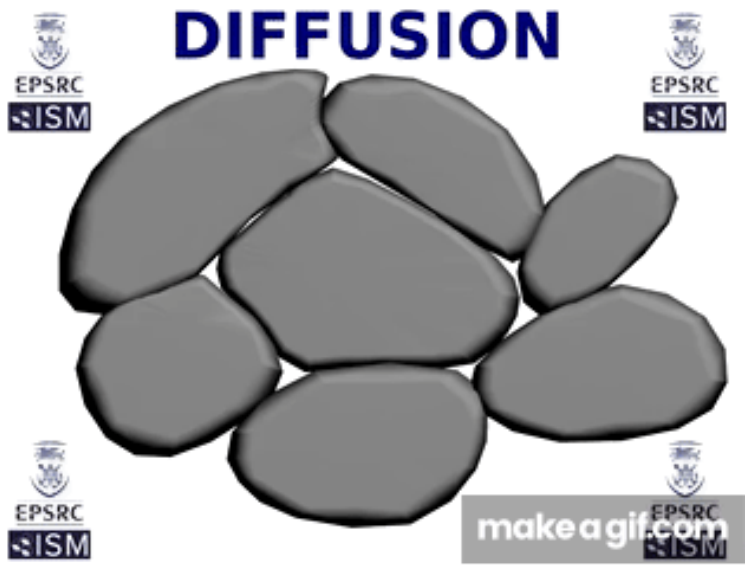
### Diffusion Creep

- Diffusion of vacancies through a crystal lattice
- Only mechanism in **Baseline simulations**

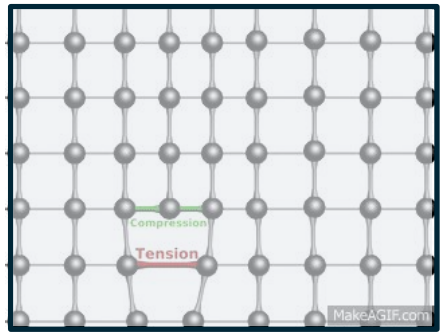
**Baseline:** $$\eta = \eta_{ref} exp\left(\frac{E_{dif}}{RT_b}\left(\frac{T_b}{T} - 1\right)\right)$$

**Non-Baseline:** $$\eta_{dif} = \frac{\dot{\epsilon}^{\frac{1-n_{dif}}{n_{dif}}} d^{\frac{m_{\mathrm{dif}}}{n_{\mathrm{dif}}}}}{A_{dif}^{-n_{dif}}} \exp\left(\frac{E_{dif} + PV_{\mathrm{dif}}}{n_{dif} RT}\right)$$

**Diffusion Creep**



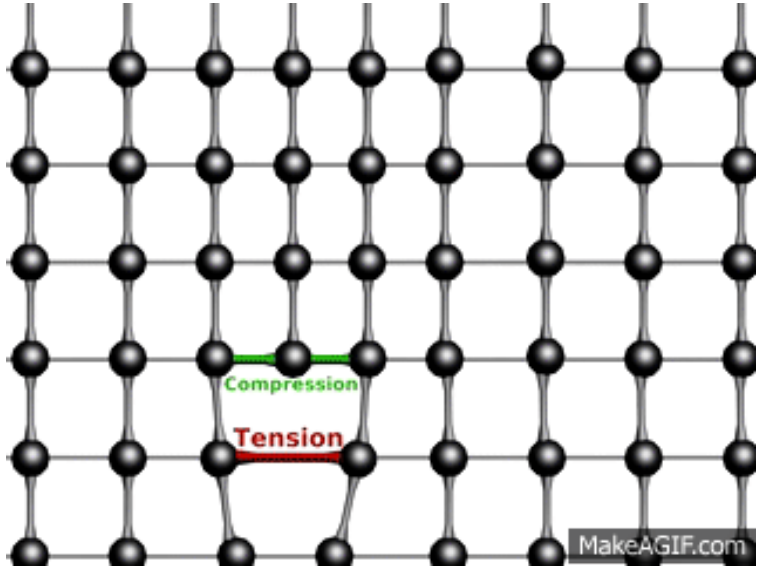

$$\frac{1}{\eta_{eff}} = \frac{1}{\eta_{diff}} + ?$$




# Dislocation
## Non-Baseline: Complex Rheology

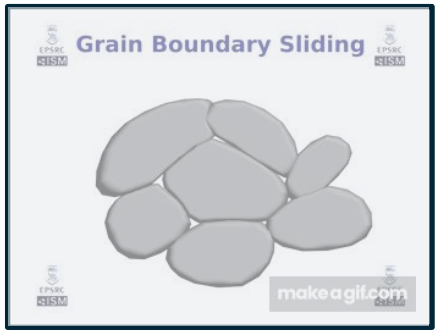
- Reminder: Deformation mechanisms determine how material flows in the shell
- Dislocation: Edges of lattice get shifted due to stress

**Dislocation Creep**




$$\eta_{dis} = \frac{\dot{\epsilon}^{\frac{1-n_{dis}}{n_{\mathrm{dis}}}} d^{\frac{m_{\mathrm{dis}}}{n_{\mathrm{dis}}}}}{A_{dis}^{-n_{dis}}} \exp\left(\frac{E_{dis} + PV_{\mathrm{dis}}}{n_{dis} RT}\right)$$

$$\frac{1}{\eta_{eff}} = \frac{1}{\eta_{diff}} + \frac{1}{\eta_{disl}} + ?$$

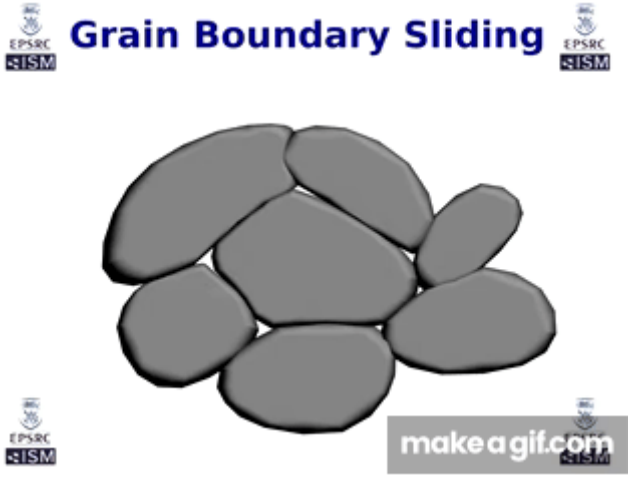



# Grain Boundary Sliding
## Non-Baseline: Complex Rheology

- Not frequently simulated for silicate bodies due to grain size differences

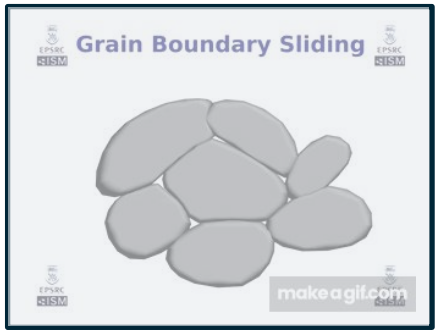
- Mantle Olivine usually cm's or mm's, while ice grains usually > 1 mm

**Grain Boundary Sliding**




$$\eta_{\mathrm{gbs}} = \frac{\dot{\epsilon}^{\frac{1-n_{\mathrm{gbs}}}{n_{\mathrm{gbs}}}} d^{\frac{m_{\mathrm{gbs}}}{n_{\mathrm{gbs}}}}}{A_{\mathrm{gbs}}^{-n_{\mathrm{gbs}}}} \exp\left(\frac{E_{\mathrm{gbs}} + PV_{\mathrm{gbs}}}{n_{\mathrm{gbs}} RT}\right)$$

$$\frac{1}{\eta_{eff}} = \frac{1}{\eta_{diff}} + \frac{1}{\eta_{disl}} + \frac{1}{\eta_{\mathrm{gbs}}} + ?$$




# Basal Slip
## Non-Baseline: Complex Rheology

- Ice 1h has extreme **plastic anisotropic**

- Olive's crystalline structure doesn't enable the same anisotropic deformation

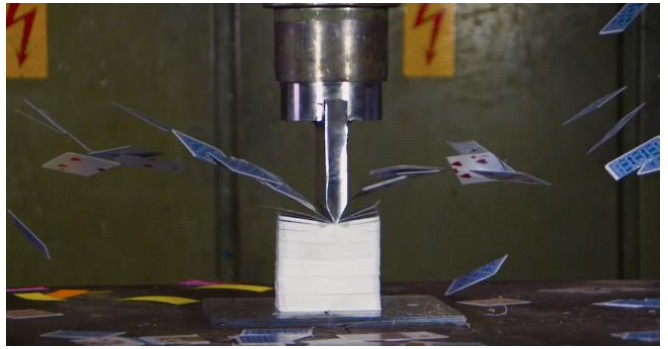
$$\eta_{\mathrm{bs}} = \frac{\dot{\epsilon}^{\frac{1-n_{\mathrm{bs}}}{n_{\mathrm{bs}}}} d^{\frac{m_{\mathrm{bs}}}{n_{\mathrm{bs}}}}}{A_{\mathrm{bs}}^{-n_{\mathrm{bs}}}} \exp\left(\frac{E_{\mathrm{bs}} + PV_{\mathrm{bs}}}{n_{\mathrm{bs}} RT}\right)$$

$$\frac{1}{\eta_{eff}} = \frac{1}{\eta_{diff}} + \frac{1}{\eta_{disl}} + \frac{1}{\eta_{\mathrm{gbs}}} + \frac{1}{\eta_{\mathrm{bs}}}$$

**Easy to deform**

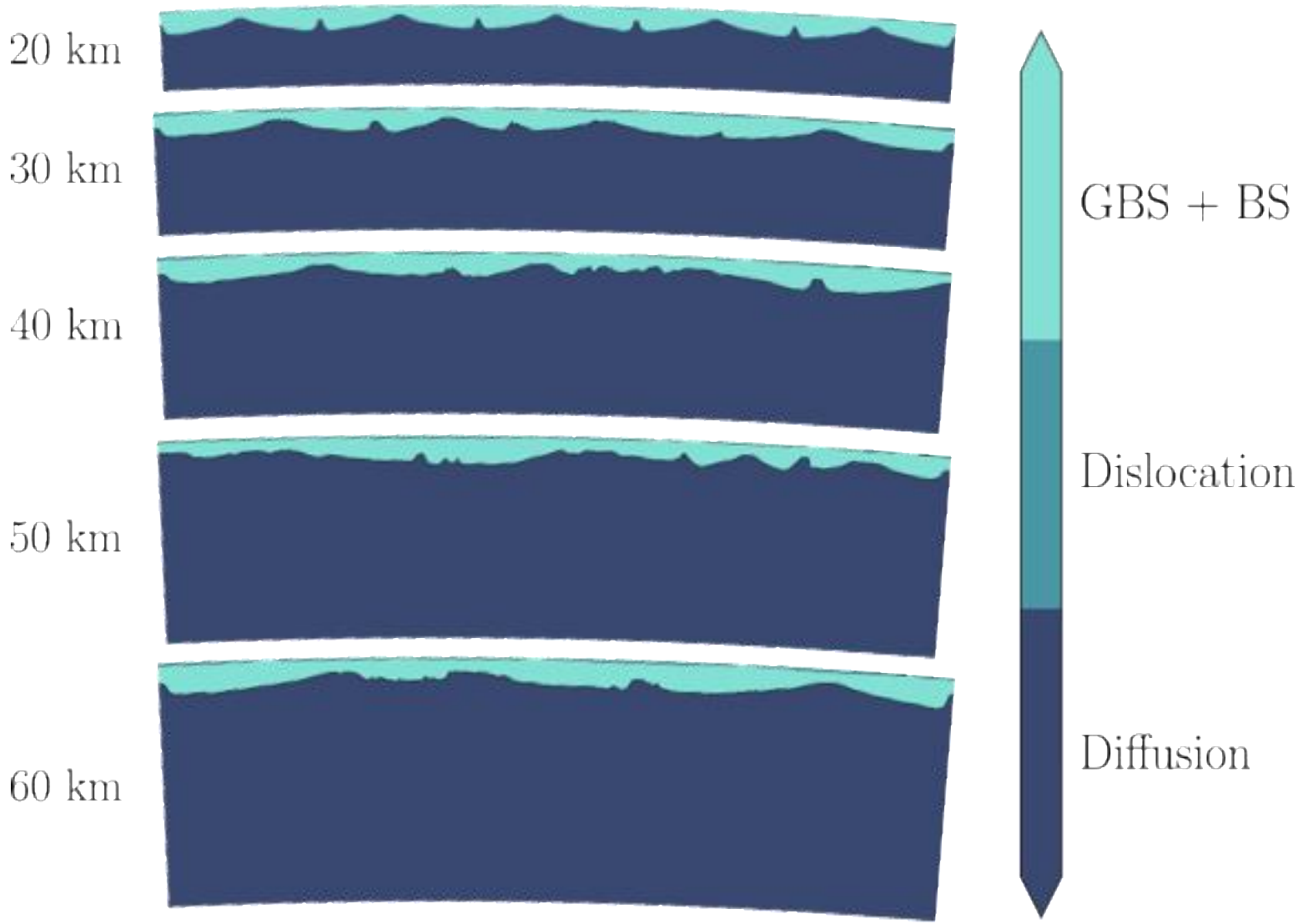



**Challenging to deform**



# 2D Deformation Field

## Non-Baseline: Complex Rheology



# Thermal Conductivity & Expansivity

## Non-Baseline: Variable K & α

- In the simple parameter study, thermal conductivity (k) and thermal expansivity (α) are both kept constant depth-wise
- Constant Parameterizations

$$k = k_{\text{ref}} = 2.418 \quad \frac{W}{m \cdot K}$$

**Petrenko and Whitworth (1999)**

$$\alpha = \alpha_{\text{ref}} = 1.492 \cdot 10^{-4}\ \text{K}^{-1}$$

**Feistel and Wagner (2006)**

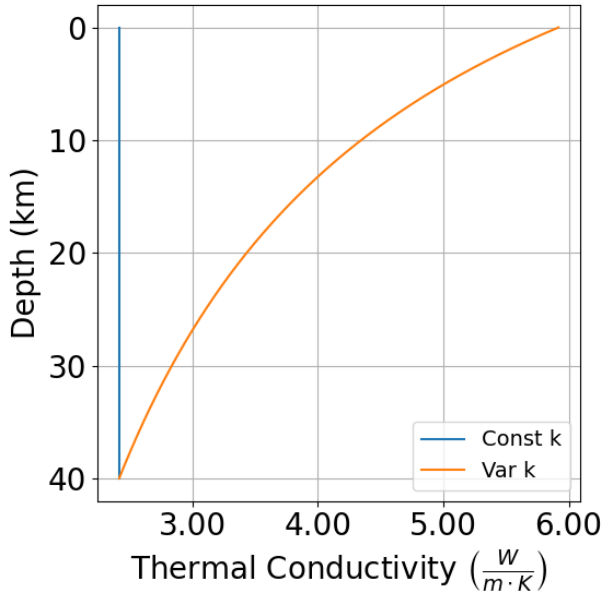
- Variable Parameterizations

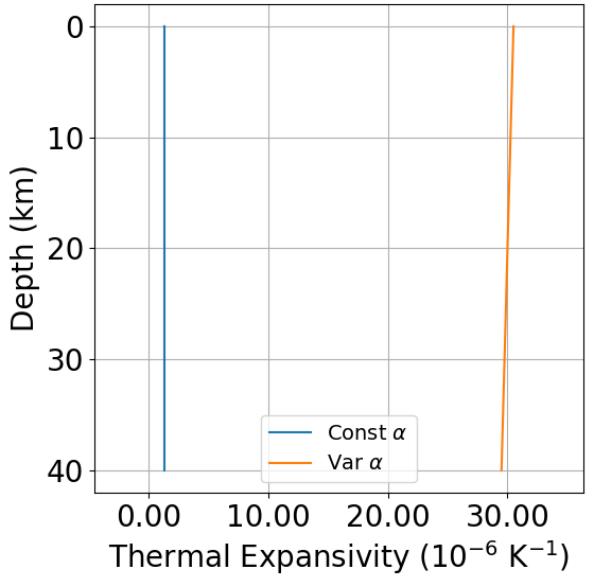
$$k(T) = \frac{651}{T}$$

**Petrenko and Whitworth (1999)**

$$\alpha(T, P) = A(T) + B(T) \cdot P$$

**Rückriemen-Bez and Plesa (2025)**



# Radar Attenuation Electrical Conductivity

## Lateral Detectability

| | | | |
|---|---|---|---|
| $E_a$ | activation energy of ice viscosity[b] | 2.2, 50, 60 | kJ mol$^{-1}$ |
| $T_r$ | reference temperature[e] | 251 | K |
| $k_B$ | Boltzmann constant | $8.6173 \times 10^{-5}$ | eV K$^{-1}$ |
| $\sigma_0^0$ | electrical conductivity of pure ice[e] | 9.2 | μS m$^{-1}$ |
| $C_0$ | auxiliary constant[e] | 1 | - |
| $E_0$ | activation energy of pure ice[e] | 0.51 | eV |
| $\sigma_1^0$ | molar electrical conductivity of acid ice[e] | 3.2 | S m$^{-1}$ M$^{-1}$ |
| $C_1$ | radar effective molar concentration of acid[e] | 1.2 | μM |
| $E_1$ | activation energy of acid ice[e] | 0.20 | eV |
| $\sigma_2^0$ | molar electrical conductivity of salty ice[e] | 0.43 | S m$^{-1}$ M$^{-1}$ |
| $C_2$ | radar effective molar concentration of salt[e] | 4.1 | μM |
| $E_2$ | activation energy of salty ice[e] | 0.19 | eV |

**Kalousová et al., 2017**

$$\sigma = \sum_{i=0}^{2} \sigma_i C_i \exp\left[-\frac{E_i}{k_b}\left(\frac{1}{T} - \frac{1}{T_r}\right)\right]$$

$$A_2(d) = \sum_{i=0}^{i_d} 2A_1(d_i)\Delta d$$

$$A_1 \approx 0.914\,\sigma$$

**Low Loss:** Pure Water Ice (i = 0)

$$A_1 = 0.914\,\sigma_0$$

**High Loss:** Water ice, salts, acids (i = 0,1,2)

$$A_1 = 0.914\,(\sigma_0 + \sigma_1 + \sigma_2)$$

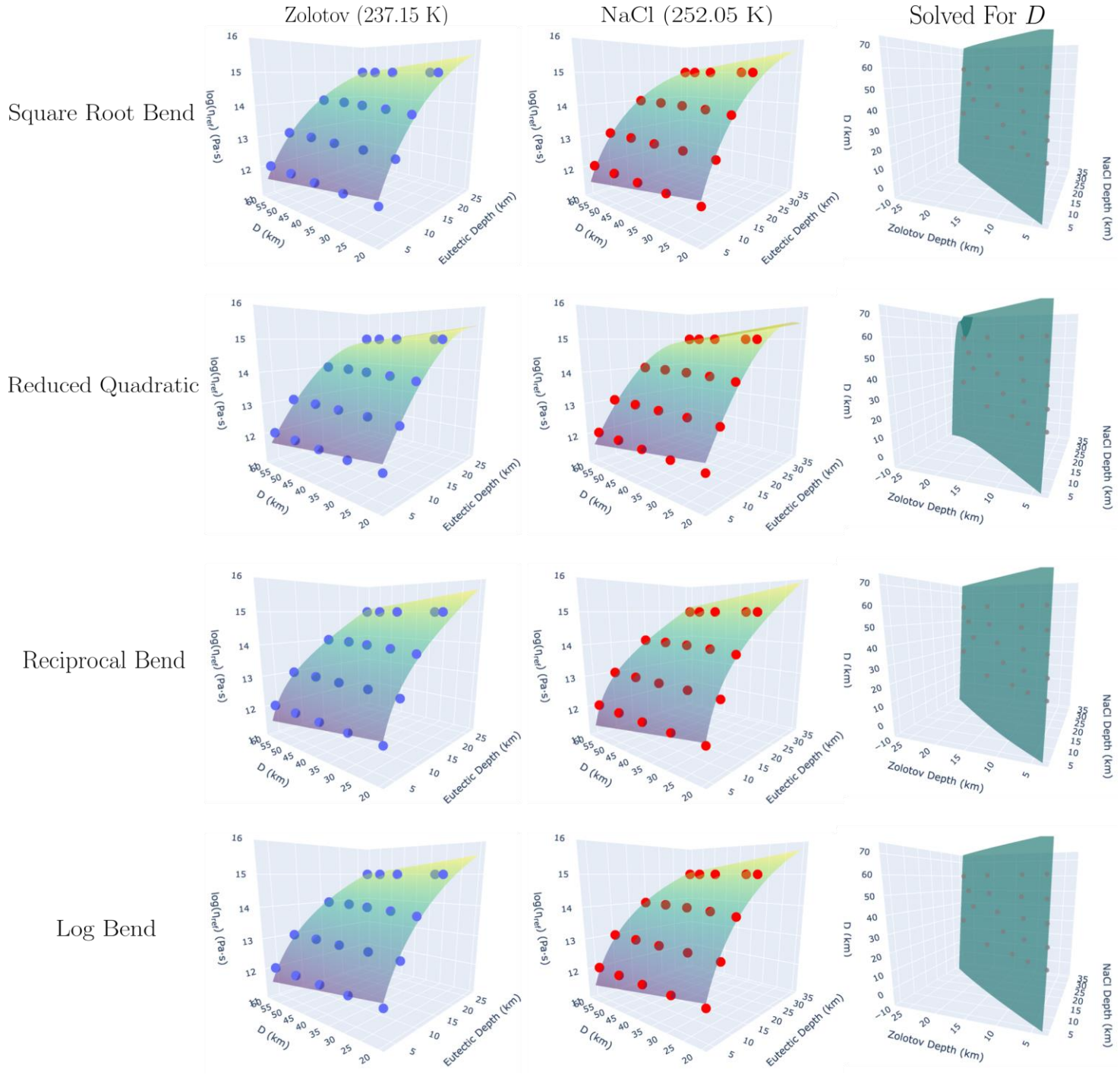
Compounds mentioned explicitly in **MacGregor et al. 2007**

**Acids** ($H^+$)
- $H_2SO_4$, HCl, $HNO_3$, $CH_3SO_3H$

**Salts** ($Cl^-$)
- NaCl

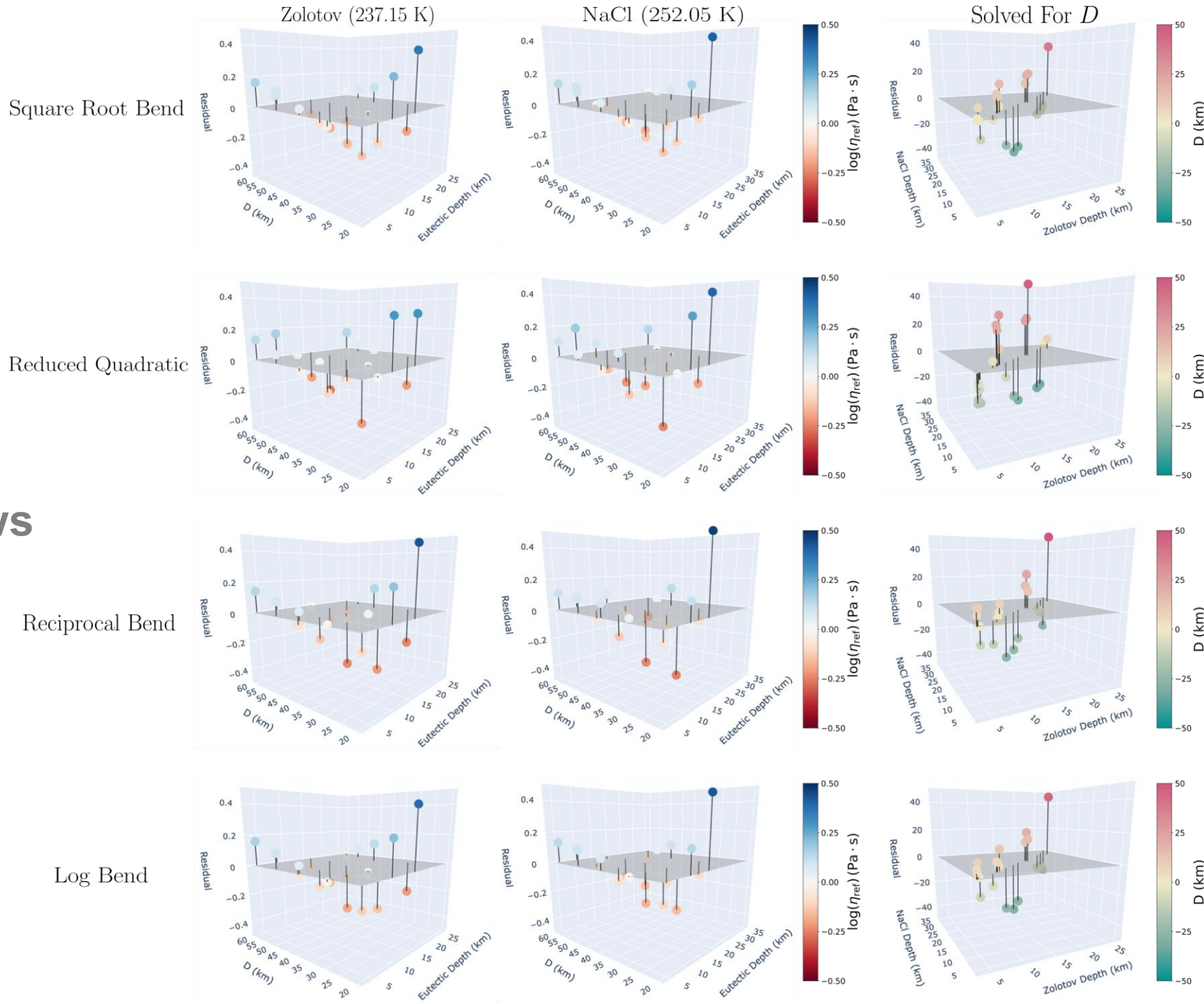
# Solution Surfaces: D

## Geophysical Scaling Laws



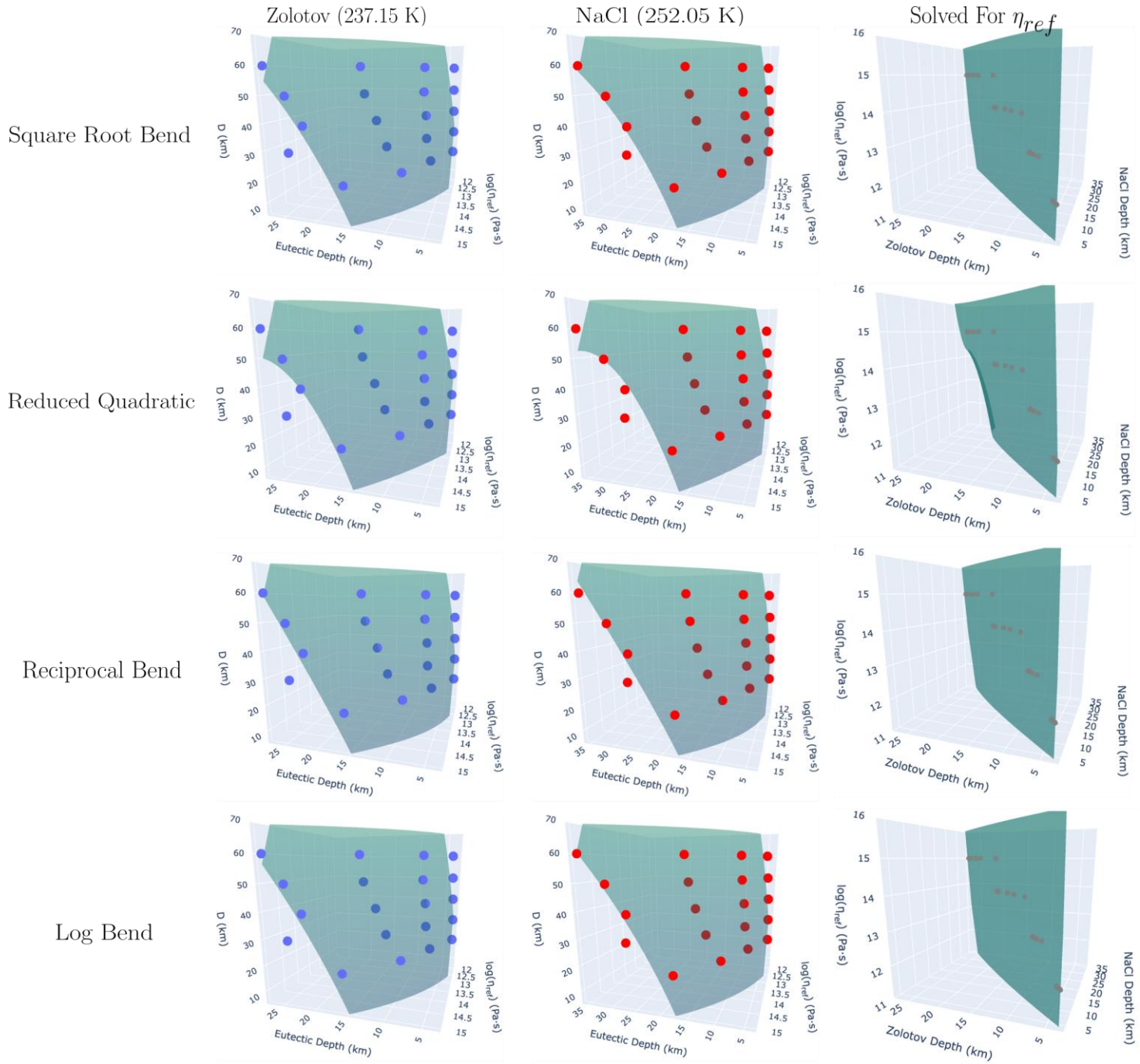
# Residuals: D

## Geophysical Scaling Laws

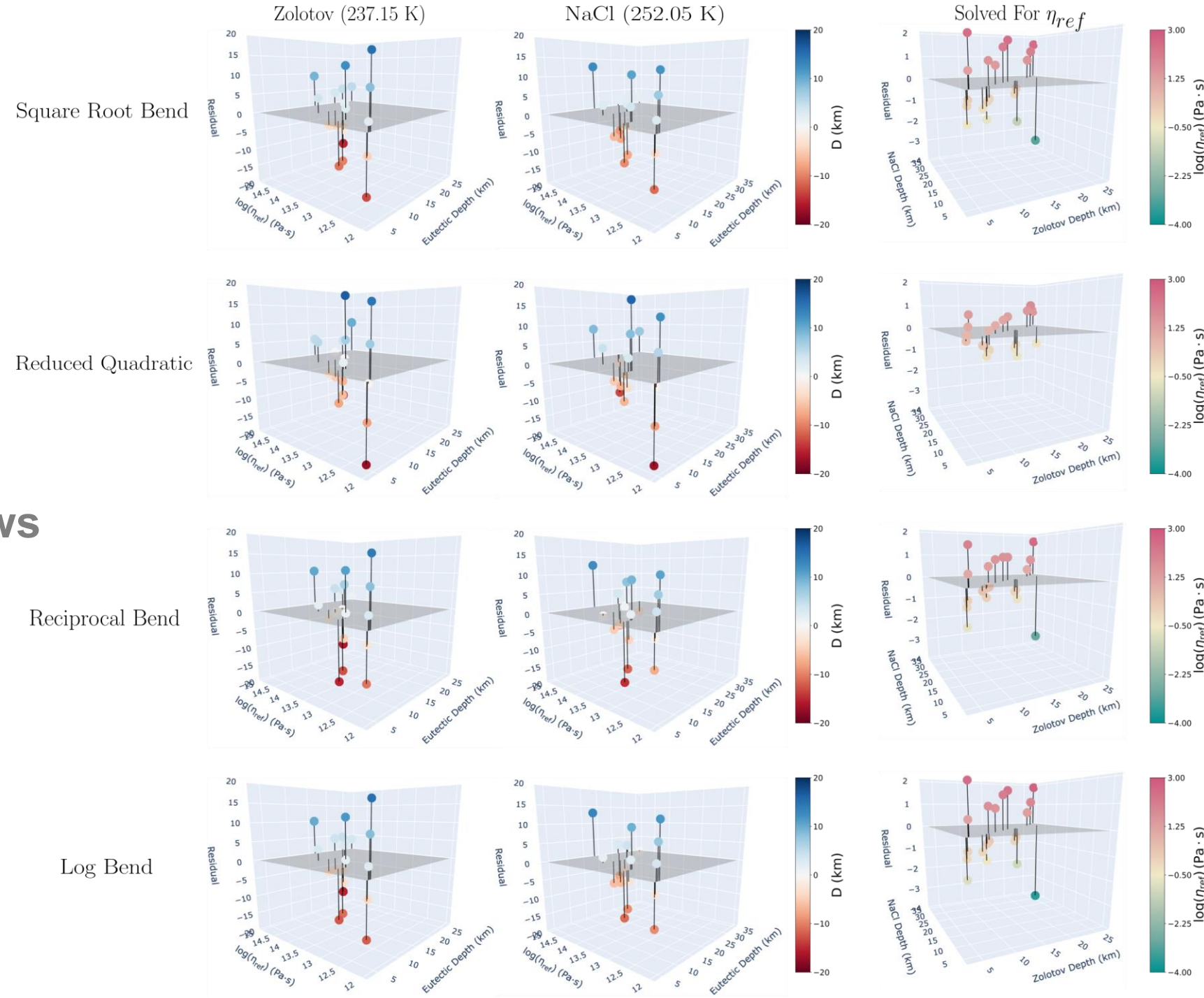


# Solution Surfaces: $\eta_{ref}$

## Geophysical Scaling Laws

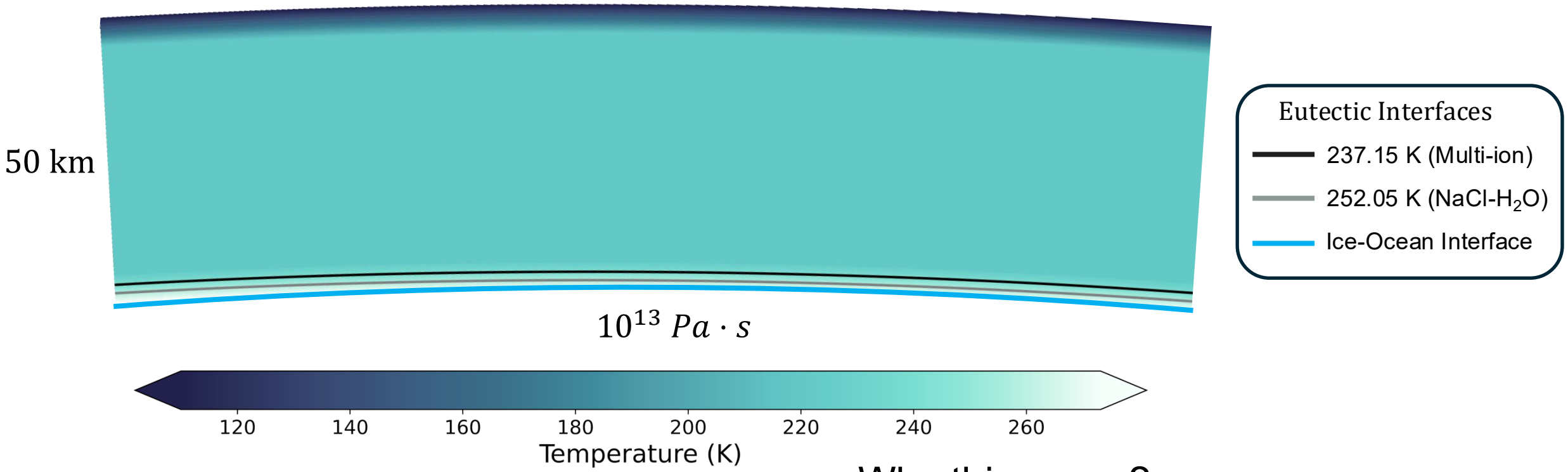


# Residuals: $\eta_{ref}$
## Geophysical Scaling Laws



# Example Thermal Evolution
## Simulations

*Ends at ~10 Myr




## Baseline Parametric Variation

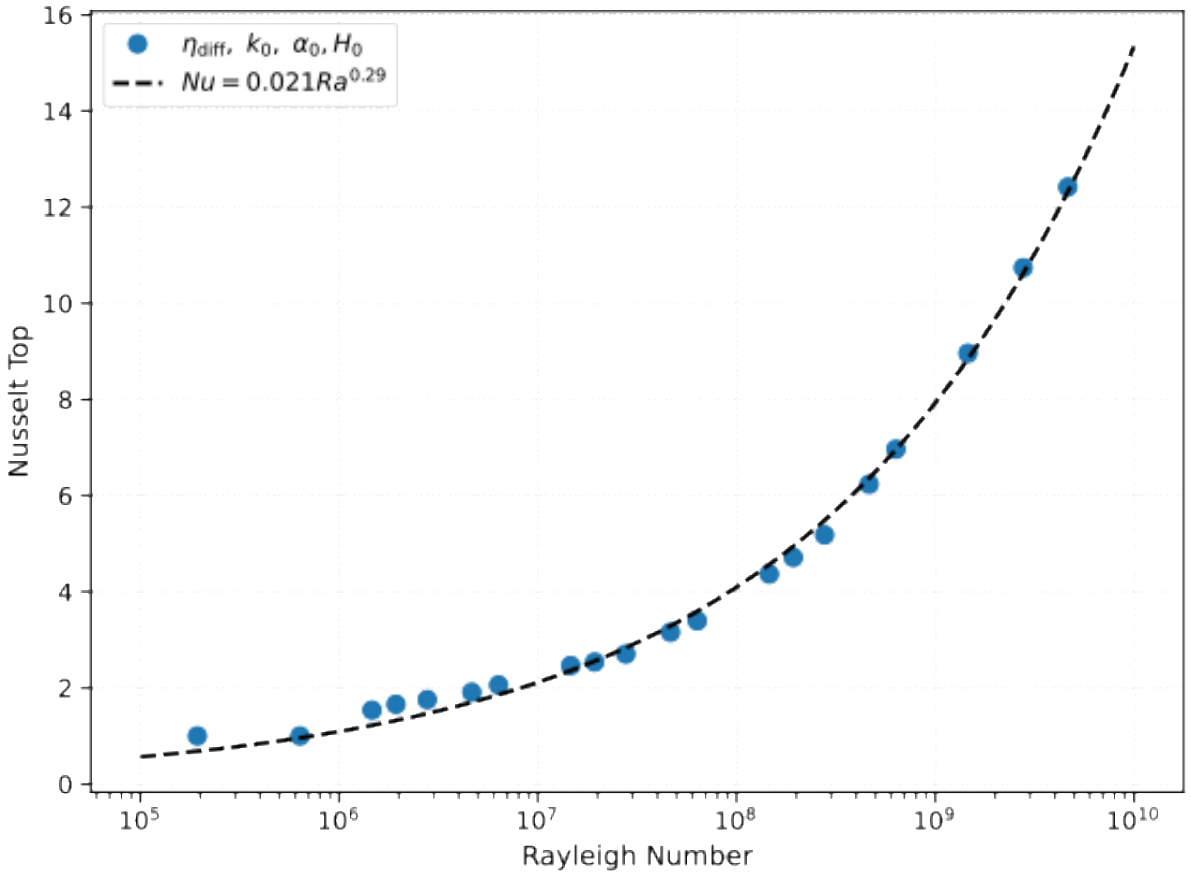
- Ice Shell Thickness (D): 20 - 60 km
- Reference Viscosity ($\eta_{ref}$): $10^{12}$–$10^{15}\ Pa \cdot s$

## Why this range?

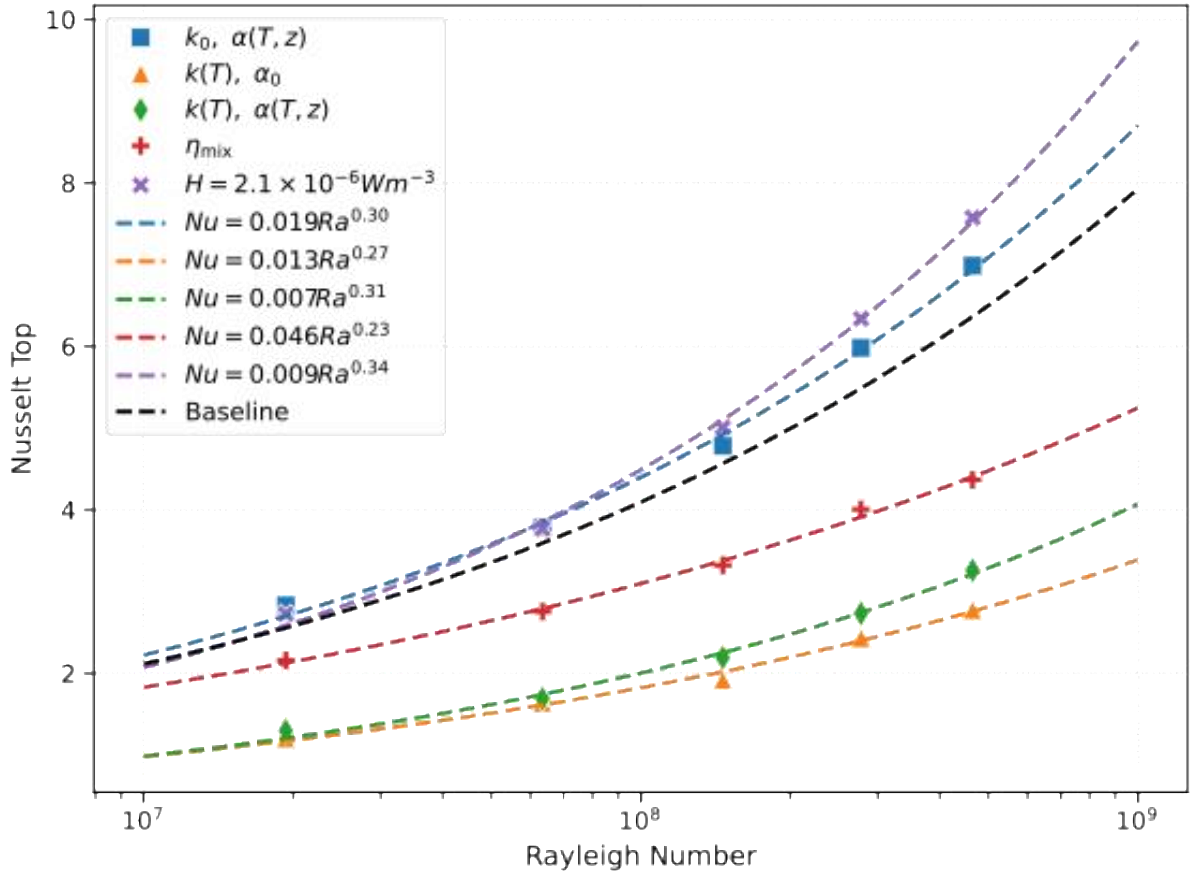
- Poorly constrained parameters
- Covers all possible convective regimes
- Effects isotherm depth & geometry

# Nusselt Rayleigh Scaling
## Geodynamic Scaling Laws

### Baseline




### Non-Baseline



# Conclusion

- Eutectic **interface detectability** remains **possible across our entire parameter study**, with a notable **decrease** laterally for **sluggishly convective** simulations of up to **34.1%**.

- We **identified scaling relationships** between **mean eutectic depth** and **shell thickness & reference viscosity**, however our limited dataset shows a weak relation

Future Work

- Improve results by **grouping scaling laws** according to **convective regimes** and perform a higher **resolution parameter study**
- More realistic results could be achieved with **increased simulation complexity** and incorporating **additional reductions to radar sounding losses**